

An Anatomy of the Baumol Cost Disease*

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Abstract

We study how structural change shapes aggregate productivity through the allocation of production and innovation. Using U.S. data, we document that production employment moves from Goods toward non-research-intensive Services, while research employment moves predominantly toward research-intensive Services. We develop a general equilibrium model in which sectoral composition, capital deepening, and directed innovation are jointly determined, with heterogeneous markups and cross-sector knowledge spillovers. Holding initial value added shares fixed while retaining baseline sectoral productivity paths raises average annual aggregate productivity growth by 0.13 percentage points over 1950–2023. Removing the forces driving changes in value added shares and re-solving the equilibrium raises growth by 0.37 percentage points, leaving productivity approximately 31 percent higher in 2023. An exact decomposition attributes 62 percent of this growth difference to the endogenous response of sectoral productivity. Innovation provides the largest positive contribution: sustaining the Goods sector’s market share supports research and idea accumulation, reinforced by its reliance on its own knowledge stock. These findings show that the productivity consequences of structural change depend on both the reallocation of production and the innovation incentives generated by that reallocation.

Keywords: Baumol’s cost disease; structural change; directed technical change; endogenous innovation; knowledge spillovers.

JEL codes: O11, O14, O33, O41.

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1 Introduction

A defining feature of post-industrial economies is the sustained reallocation of economic activity from goods-producing sectors, where productivity growth is relatively fast, toward services, where it is slower. A central explanation for the productivity consequences of this reallocation is Baumol’s cost disease: when productivity grows more slowly in the sectors absorbing production activity, changes in sectoral composition can reduce aggregate productivity growth (Baumol 1967, Baumol et al. 1985). Standard treatments of this mechanism, however, typically take sectoral productivity growth as given.¹ This abstracts from a potentially important feedback. Differences in sectoral productivity growth contribute to structural change through their effects on relative prices, while the sectoral composition of the economy shapes market sizes and therefore the incentives to develop technologies for different sectors.

This paper studies that feedback. Our starting point is that directed technical change provides a natural framework for understanding how sectoral composition can shape productivity dynamics. In these models, innovation is directed toward technologies with higher expected returns, which depend on the size of the market they serve and on relative prices (Acemoglu 2002). The sectoral composition of the economy therefore affects not only the allocation of production, but also the sectors in which research is undertaken and, through innovation, the evolution of sectoral productivity growth. This mechanism suggests that changes in the sectoral composition of production should be accompanied by corresponding changes in the direction of innovative activity.

The data reveal that this connection between the composition of production and the direction of innovation is more nuanced than the standard benchmark suggests. We document this pattern using postwar U.S. data that combine industry-level R&D employment and patents with measures of production, value added, prices, labor productivity, and capital. A key feature of the data is the substantial heterogeneity in innovative activity within Services. Some service industries are as research intensive as Goods, while others display very little research activity. This motivates our division of the economy into three broad sectors: Goods, research-intensive services (RI-Services), and non-research-intensive services (NRI-Services). Both service sectors expand as Goods declines, but NRI-Services remains by far the larger sector and absorbs a substantial share of the expansion in production activity. The reallocation of research looks different: research employment shifts predominantly toward RI-Services, while NRI-Services attracts comparatively little of the expansion in innovative activity. The service industries that account for most production activity are therefore not the same industries toward which research is increasingly

¹Some examples of this include McMillan et al. (2014), Timmer & Szirmai (2000) and De Vries et al. (2015).

directed. These differences in the direction of innovation are accompanied by substantial differences in productivity growth, with labor productivity growing faster in Goods and RI-Services than in NRI-Services.

To understand the mechanisms governing the joint endogenous allocation of production and innovation across sectors, we develop a general equilibrium model of structural change and directed technical change. Non-homothetic preferences generate sectoral reallocation through income and relative-price effects. Within each sector, production uses labor and differentiated machines, while innovation expands the set of sector-specific machine varieties. The returns to innovation depend on sectoral market size and relative prices, so changes in the composition of the economy affect the incentives to direct research across sectors. Research productivity also depends on accumulated knowledge produced both within the sector and elsewhere in the economy. The model therefore provides a quantitative laboratory for studying the feedback between sectoral production, innovation, productivity growth, and relative prices. We additionally allow final-producer and machine-producer markups to differ across sectors and evolve over time, allowing market power in final and intermediate production to shape the profitability of innovation and the sectoral allocation of research.

We discipline the model using U.S. data over 1950–2023. The calibration uses the observed evolution of sectoral value-added shares, researcher shares, relative prices, labor productivity, markups, capital prices, and cross-sector patent citations. The model delivers an exact analytical expression for sectoral labor productivity, linking its evolution to the accumulation of machine varieties, machine productivity, relative prices, markups, and labor efficiency. Crucially, the accumulation of machine varieties is endogenous and responds to the sectoral allocation and productivity of research as well as to access to accumulated knowledge. This structure allows us to distinguish the standard compositional effect of structural change from its equilibrium effect on the sectoral productivity paths themselves.

We use the calibrated model to quantify the forces driving sectoral productivity growth and to assess how structural change shapes those forces in equilibrium. Our first set of quantitative results provides an anatomy of sectoral productivity growth. The model attributes the productivity-growth difference between Goods and NRI-Services primarily to endogenous innovation. Goods accumulates machine varieties at a higher rate than NRI-Services. RI-Services reaches a labor-productivity growth rate close to that of Goods through a different combination of forces: endogenous innovation contributes less than in Goods, while embodied technical progress contributes more. Similar observed productivity growth across Goods and RI-Services therefore masks different underlying sources of technical change.

We next examine why innovation evolves so differently across sectors. The production of new ideas reflects research productivity, the allocation of researchers, accumulated knowledge, and changes in the structure of cross-sector knowledge spillovers. Goods innovation relies heavily on its own accumulated knowledge. RI-Services benefits both from the reallocation of researchers toward the sector and from knowledge produced elsewhere. NRI-Services also receives additional researchers and knowledge from other sectors, but these forces are offset by a declining path of research productivity. The slower accumulation of varieties in NRI-Services therefore does not simply reflect a lack of research employment.

Our main counterfactual asks how structural change affects aggregate labor productivity once these innovation responses are allowed to operate. We first consider the conventional fixed-share calculation, which applies 1950 value-added shares to the baseline paths of sectoral productivity. Under this benchmark, aggregate productivity in 2023 is about 10 percent higher than in the baseline. This captures the standard compositional effect while holding sectoral productivity paths fixed.

We then remove the forces generating changes in sectoral value-added shares and resolve the equilibrium. Specifically, we eliminate income effects, remove the response of expenditure shares to relative prices, and hold both final-producer and machine-producer markups at their 1950 values. Under this full shutdown, aggregate productivity in 2023 is approximately 31 percent higher than in the baseline, compared with about 10 percent under the fixed-share benchmark. The equilibrium effect is therefore about 21 percentage points larger, reflecting the additional equilibrium adjustments induced by the shutdown, including the endogenous response of sectoral productivity.

Our decomposition attributes approximately 62 percent of the 26.9 log-point increase in cumulative productivity growth under the full shutdown to the endogenous response of sectoral productivity. Endogenous innovation provides the largest positive contribution to this response. Under the full shutdown, Goods retains a larger share of economic activity, which raises the return to developing Goods-specific machines and redirects researchers toward the sector. The resulting accumulation of varieties is reinforced by Goods' reliance on its own accumulated knowledge. By 2023, the stock of Goods machine varieties is approximately 56 percent higher than in the baseline and Goods labor productivity is approximately 66 percent higher.

These results provide a broader interpretation of the Baumol cost disease. The productivity consequences of structural change do not arise only because production is reallocated toward sectors with slower predetermined productivity growth. The reallocation of production also changes the incentives determining where innovation occurs and, through innovation, the subsequent evolution of sectoral productivity itself. In the calibrated

model, the divergence between the sectoral allocation of production and research creates an additional channel through which structural change affects aggregate productivity.

Related Literature. Our paper contributes first to the literature on structural change and aggregate productivity. This literature originates with Baumol’s cost disease ([Baumol 1967](#), [Baumol et al. 1985](#)) and studies how differences in demand and sectoral productivity generate the reallocation of economic activity across sectors. Non-homothetic demand plays a central role in [Kongsamut et al. \(2001\)](#), while differences in sectoral productivity growth generate structural change in [Ngai & Pissarides \(2007\)](#), and differences in factor intensities provide an additional supply-side mechanism in [Acemoglu & Guerrieri \(2008\)](#). More recent quantitative frameworks combine income and relative-price effects ([Boppart 2014](#), [Comin et al. 2021](#)). See also [Herrendorf et al. \(2013, 2014\)](#) for evidence and a general framework organizing these mechanisms. Closely related to our focus on productivity, [Duarte & Restuccia \(2010\)](#) study how sectoral labor-productivity differences and the reallocation of labor shape aggregate productivity across countries, while [Herrendorf et al. \(2015\)](#) emphasize differences in sectoral technical progress as a force behind U.S. structural transformation.

[Duernecker et al. \(2024\)](#) study structural change within Services and its implications for the future of Baumol’s cost disease. They find that broad service categories are substitutes, so that services with slower productivity growth do not eventually absorb all economic activity. Our sectoral aggregation instead implies strong complementarity among Goods, RI-Services, and NRI-Services. We also ask a different quantitative question. Rather than projecting the future evolution of the cost disease, we use the postwar U.S. experience to quantify the mechanisms through which structural change has affected labor productivity. In particular, sectoral productivity growth is an equilibrium outcome in our framework, allowing us to distinguish the compositional effect of reallocating production from the induced response of sectoral productivity.

The paper closest to ours is [Comin et al. \(2025\)](#), who document the structural transformation of innovation and develop a multi-sector semi-endogenous growth model in which non-homothetic demand, heterogeneous innovation technologies, and knowledge spillovers jointly determine the sectoral allocation of production and innovation. Our analysis differs in both its empirical focus and its quantitative objective. We focus on heterogeneity within Services and document that the expansion of production and the reallocation of research need not have the same destination: both service sectors expand in value added, NRI-Services remains the larger sector, while research increasingly concentrates in RI-Services. Our main productivity object is labor productivity, for which the model delivers an exact analytical expression that can be decomposed into endogenous innovation, embodied technical progress, relative-price effects, and labor efficiency. The

quantitative analysis then uses this structure to study the historical productivity consequences of structural change over 1950–2023. Our production structure additionally allows final-producer and machine-producer markups to differ across sectors and evolve over time, and allows for sector-specific embodied technical progress. These margins affect both sectoral labor productivity and the profitability of innovation.

Our treatment of innovation builds on the directed technical change literature ([Acemoglu 2002, 2007](#)) and on the semi-endogenous idea-production technology of [Jones \(1995\)](#). Directed technical change makes the profitability of innovation depend on relative prices and market size, providing the mechanism through which structural change affects the direction of research. We extend the idea-production technology to multiple sectors. Research productivity can differ and evolve across sectors, and the productivity of research directed toward a sector depends on accumulated knowledge produced both within that sector and elsewhere in the economy. This structure is also related to [Bloom et al. \(2020\)](#), who document that rising research effort can coexist with declining research productivity. In our setting, these forces operate at the sector level, allowing researcher allocation, sector-specific research productivity, and access to accumulated knowledge to jointly determine the production of new ideas.

Finally, our paper relates to work on capital-embodied technical change and market power. [Caunedo & Keller \(2024\)](#) document substantial sectoral differences in capital-embodied technical change and show that they account for an important share of differences in sectoral labor-productivity growth. [Jones & Liu \(2024\)](#) develop a growth framework in which capital-embodied technological progress can generate structural transformation, while [León-Ledesma & Moro \(2020\)](#) study how the rise of Services interacts with capital accumulation and investment-specific technical change. Our framework incorporates sector-specific embodied technical progress alongside endogenous variety creation, allowing their contributions to sectoral labor-productivity growth to be separated. We also relate to [?](#), who studies how structural transformation and the expansion of Services shape the evolution of markups. Whereas his focus is on explaining markup dynamics, we discipline sector- and time-varying markups from the data and study how market power in final and intermediate production affects labor productivity and the sectoral allocation of innovation.

Outline. The remainder of the paper is organized as follows. Section [2](#) documents the empirical patterns in production, innovation, and productivity. Section [3](#) develops the model, and Section [4](#) describes its quantitative implementation. Section [5](#) presents the quantitative exercises, and Section [6](#) concludes.

2 Empirics

This section documents novel patterns of structural change in innovation in the post-war US economy. We supplement these with data on labor reallocation, labor productivity and relative prices at the sector-level to motivate our theory of structural change with endogenous sectoral productivity growth.

2.1 Data Sources

We combine four publicly available data sources covering production, employment, patents, and patent citations. We begin with two publicly available sources of industry-level production data. The first is the US KLEMS March 2017 release ([Jorgenson et al. 2017](#)). The second is the BEA–BLS Integrated Industry-Level Production Accounts, for which we use both the historical 1947–2016 accounts ([Eldridge et al. 2020](#)) and the most recent release covering 44 industries over 1947–2023 and 63 industries over 1963–2023 ([Garner et al. 2025](#)). We use these data to construct sector-level series for value added shares, relative prices, and labor productivity over the postwar period.²

We complement the production data with a new occupation-based measure of innovation inputs by industry. We construct this measure using IPUMS USA ([Ruggles et al. 2025](#)), which provides harmonized microdata on workers’ occupations and industries over a long time span. We begin from the set of STEM occupations classified by O*NET as “Research, Development, Design and Practitioners” or “Research, Development, Design and Practitioners; Technologists and Technicians,” and retain those occupations that correspond to R&D activity relevant for the creation and development of new knowledge and technologies. We then map these occupations into the harmonized occupation classifications available in IPUMS and assign workers to industries using consistent industry crosswalks across census years. Aggregating person-level observations with the corresponding survey weights yields an industry-year panel of R&D employment spanning 1950–2023.³

Our third source is the US Patent and Trademark Office Historical Patent Annual Data File ([Marco et al. 2015](#)), which provides patent-level information from 1840 to 2014. Each patent is assigned a United States Patent Classification (USPC) code, which we map into

²Combining these sources into a consistent long-run panel requires harmonizing industry definitions and coverage across releases. Some of the production and productivity measures required for our analysis are also not reported directly in both sources. We therefore construct consistent nominal and real measures of output and inputs, the corresponding quantity indices, and measures of labor productivity following the US KLEMS production-account methodology in [Jorgenson et al. \(2012\)](#). The full harmonization procedure, construction of the productivity measures, and replication package are available at [this Github repository](#). Appendix A provides further details on the industry coverage used in the analysis.

³Appendix B describes the IPUMS samples, occupational screening, occupation and industry crosswalks, and construction of the industry-level R&D employment series.

NAICS industries using the probabilistic crosswalk developed by [Goldschlag et al. \(2016\)](#). Aggregating the resulting industry assignments allows us to construct annual industry-level measures of both patent flows and patent stocks, which we use as measures of innovation output over time.⁴

Our fourth source is the OECD Citations Database ([Webb et al. 2005](#)), which we use to measure citation-based knowledge linkages across sectors. We restrict the sample to citations made by patents granted by the US Patent and Trademark Office and merge these citation links with patent technology classifications from PatentsView.⁵ We then map technology classes into ISIC Rev. 3.1 industries using the concordance of [Goldschlag et al. \(2016\)](#) and aggregate the resulting citation links to our three broad sectors. This allows us to construct within- and between-sector citation probabilities over 1976–2020.⁶

2.2 Facts

We begin by documenting a broad reallocation of innovation activity from Goods toward Services.⁷ Figure 1 shows that this pattern appears in both the sectoral allocation of R&D workers, our measure of innovation inputs, and the sectoral distribution of patents, our measure of innovation output. The Goods share of R&D workers declines from 60.6% in 1950 to 23.5% in 2023. A similar pattern appears in patents, with the Goods share of issued US patents falling from 94.8% in 1950 to 78.7% in 2014. Innovation activity has therefore shifted substantially toward Services along both the input and output margins.

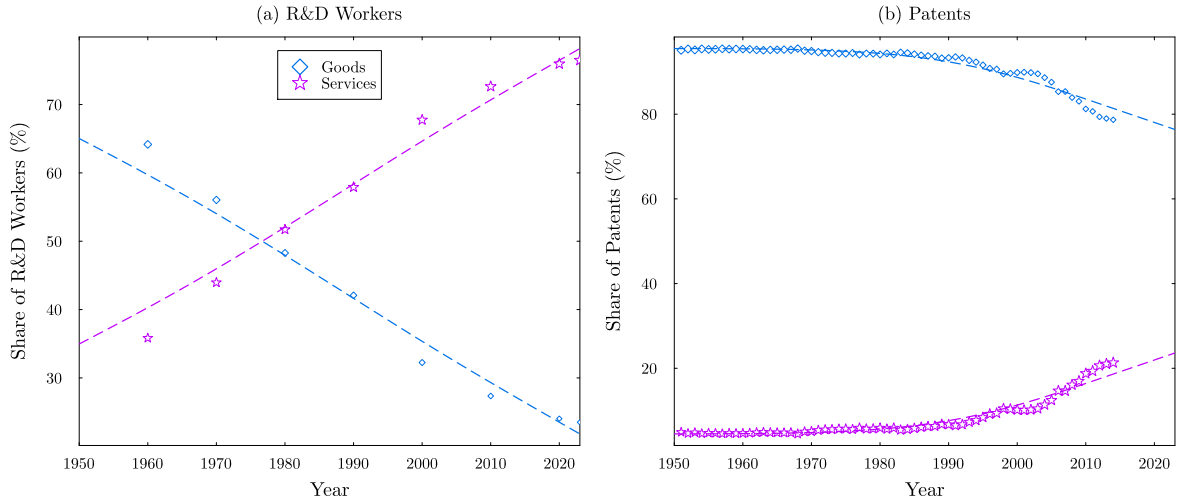
⁴Appendix C describes the patent-to-industry mapping and the construction of the industry-level patent series.

⁵We obtain the PatentsView data from the final release of the pre-grant and granted patent metadata archived by the US Patent and Trademark Office on Zenodo ([United States Patent and Trademark Office 2025](#)).

⁶Appendix D describes the construction of the citation matrix, including the treatment of patents assigned to multiple technology classes, the mapping of cited applications to US patent equivalents, and the aggregation from technologies to sectors. The sample begins in 1976, the first year for which PatentsView reports the required technology classifications, and ends in 2020 to limit truncation in citations for patents granted near the end of the sample.

⁷We aggregate industries into two broad sectors following Table A.1. Goods includes Agriculture, Construction, Mining, Utilities, and Manufacturing, corresponding to NAICS codes beginning with 1, 2, or 3, while Services includes the remaining industries.

Figure 1. Structural Change in Innovation



Notes: The left panel reports the shares of R&D workers employed in Goods and Services, constructed from IPUMS USA (Ruggles et al. 2025); see Appendix B for details. The right panel reports the corresponding shares of issued U.S. patents from the USPTO Historical Patent Annual Data File (Marco et al. 2015); see Appendix C for the patent-to-industry mapping. Goods comprises industries with NAICS codes beginning with 1–3, while Services comprises industries with NAICS codes beginning with 4–9. Hollow diamonds denote Goods observations, and hollow stars denote Services observations. Marker area is proportional to the corresponding sector’s value-added share. Dashed lines are cubic smoothing splines with smoothing parameter 10^4 .

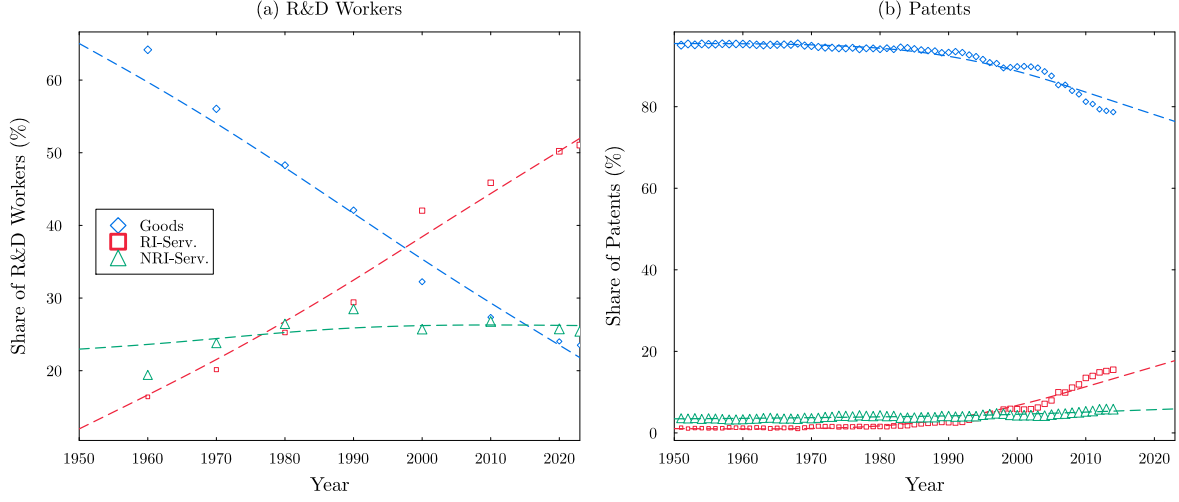
The aggregate shift of innovation activity toward Services masks substantial heterogeneity within the sector. We therefore split Services into Research-Intensive Services (RI-Services) and Non-Research-Intensive Services (NRI-Services). We classify service industries according to their average R&D-worker intensity over 1970–2020, using the corresponding average for Manufacturing as the cutoff. This procedure places Information, Finance and Insurance, and Professional, Scientific and Technical Services in RI-Services, with all remaining service industries classified as NRI-Services.⁸

Figure 2 shows that almost the entire reallocation of innovation activity toward Services is accounted for by RI-Services. The share of all R&D workers employed in RI-Services rises from 14.1% in 1950 to 51.1% in 2023, an increase of 37 percentage points that closely mirrors the decline in the Goods share over the same period. The patent evidence displays the same qualitative pattern: the RI-Services share of issued patents rises from 1.4% in 1950 to 15.5% in 2014 and overtakes the NRI-Services share by the mid-1990s. By contrast, the NRI-Services share of innovation activity remains broadly flat, increasing by only 0.2 percentage points for R&D workers and 2.1 percentage points for patents over the corresponding sample periods. Treating Services as a single sector therefore conceals

⁸R&D intensity is measured using the industry-level R&D employment series constructed from IPUMS. Appendix E describes the construction of the intensity measure, the choice of the Manufacturing benchmark, and the resulting industry classification in detail.

a sharp divergence in the allocation of innovation activity within Services.

Figure 2. Structural Change in Innovation: Heterogeneity Within Services

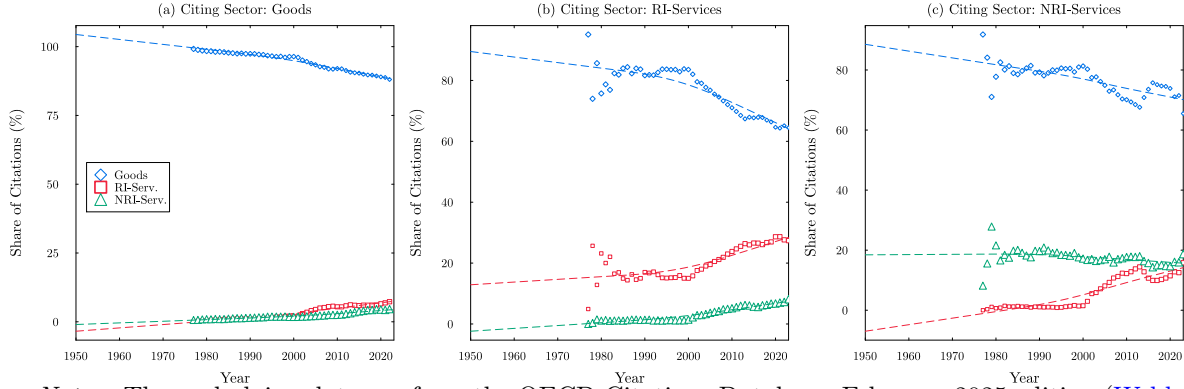


Notes: The left panel reports the sectoral shares of R&D workers constructed from IPUMS USA (Ruggles et al. 2025), while the right panel reports the sectoral shares of issued U.S. patents from the USPTO Historical Patent Annual Data File (Marco et al. 2015). RI-Services includes Information, Finance and Insurance, and Professional, Scientific and Technical Services; NRI-Services includes all remaining service industries. Appendix E describes the R&D-intensity measure and the classification of service industries. Goods comprises industries with NAICS codes beginning with 1–3. Hollow diamonds, squares, and triangles denote observations for Goods, RI-Services, and NRI-Services, respectively. Marker area is proportional to the corresponding sector’s value-added share. Dashed lines are cubic smoothing splines with smoothing parameter 10^4 .

We next examine how patents in each sector draw on knowledge generated in the other sectors. For each citing sector and year, we compute the probability that a cited patent belongs to Goods, RI-Services, or NRI-Services; Figure 3 reports the resulting within- and between-sector citation shares. Goods is the most frequently cited source sector for patents in all three sectors, but its importance declines over time. At the same time, RI-Services becomes an increasingly important source of cited knowledge across sectors.⁹ The citation evidence therefore points to a changing structure of technological linkages across sectors, with RI-Services becoming progressively more important in the knowledge base on which subsequent innovation draws.

⁹These patterns should be interpreted with some caution because patenting is heavily concentrated in technologies mapped to Goods. A high citation share toward Goods therefore reflects both the size of the Goods knowledge base and the propensity of patents in other sectors to cite it. Appendix D shows that citations made by Goods patents are close to what would arise under independence between citing and cited technologies, whereas citations made by patents in both service sectors display substantially stronger selective citation patterns.

Figure 3. Citation-Based Knowledge Linkages Across Sectors



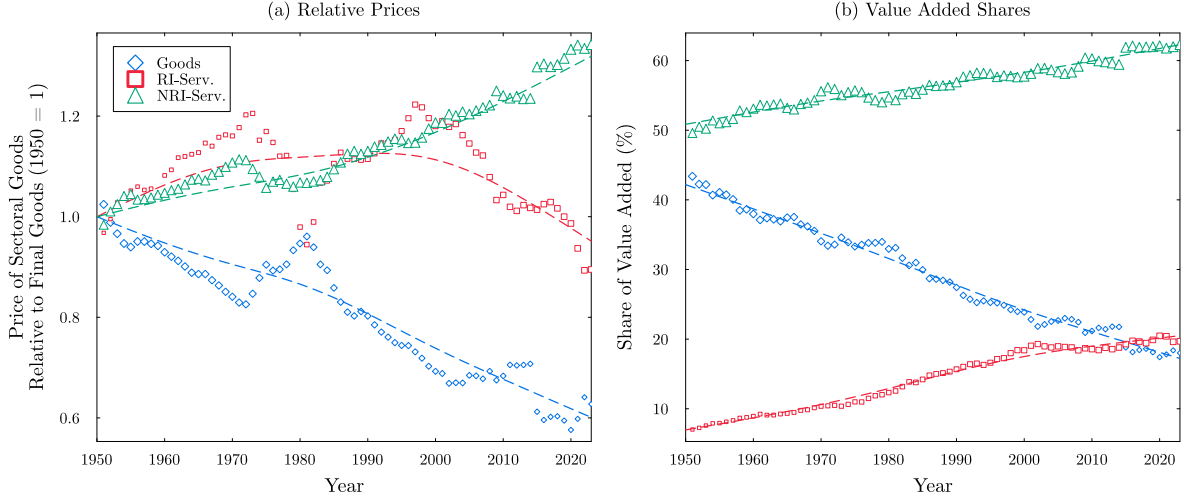
Notes: The underlying data are from the OECD Citations Database, February 2025 edition (Webb et al. 2005), restricted to citations made by patents granted by the U.S. Patent and Trademark Office between 1976 and 2020. Patent technology classes are obtained from PatentsView and mapped to industries using the concordance developed by Goldschlag et al. (2016); see Appendix D for details. Hollow diamonds, squares, and triangles denote citations to Goods, RI-Services, and NRI-Services, respectively. Marker area is proportional to the corresponding cited sector’s value-added share. Dashed lines are cubic smoothing splines with smoothing parameter 10^4 .

We next document the production-side structural change accompanying the reallocation of innovation. Figure 4 reports sectoral relative prices and value-added shares. The relative price of Goods falls by two-fifths between 1950 and 2023, while that of NRI-Services rises by almost one third.¹⁰ RI-Services displays no comparable secular trend, with its relative price fluctuating around its initial level before ending slightly below it. Over the same period, the Goods share of value added falls from 42.6% to 18.0%, while both service sectors expand.¹¹ The RI-Services share rises by 12.4 percentage points and the NRI-Services share by 12.1 percentage points, leaving NRI-Services as the largest sector at 62.3% of aggregate value added in 2023.

¹⁰Sectoral relative prices are defined as sectoral value-added price indices divided by the economy-wide value-added price index. All relative-price indices are normalized to one in 1950.

¹¹The decline in the Goods share and corresponding expansion of Services is a standard feature of structural transformation; see, for example, Herrendorf et al. (2014).

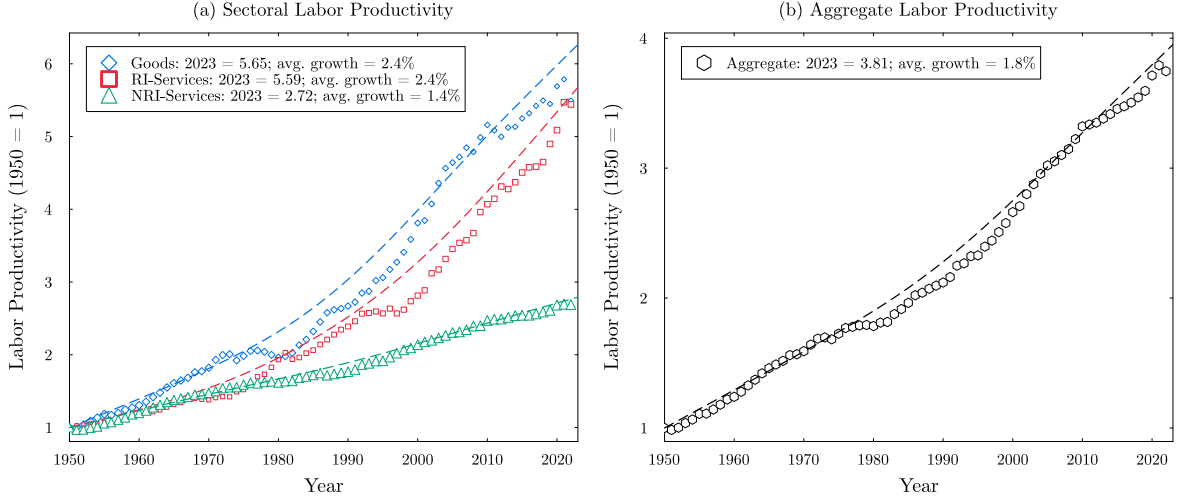
Figure 4. Relative Prices and Value-Added Shares



Notes: The left panel reports sectoral value-added price indices relative to the economy-wide value-added price index. Industry-level price indices are aggregated into the three broad sectors using Törnqvist indices and normalized to one in 1950. The right panel reports each sector's share of nominal aggregate value added. The underlying industry-level data are from U.S. KLEMS and the BEA-BLS Integrated Industry-Level Production Accounts; see Subsection 2.1 for details. RI-Services includes Information, Finance and Insurance, and Professional, Scientific and Technical Services; NRI-Services includes all remaining service industries. Hollow diamonds, squares, and triangles denote observations for Goods, RI-Services, and NRI-Services, respectively. Marker area is proportional to the corresponding sector's value-added share. Dashed lines are cubic smoothing splines with smoothing parameter 10^4 .

We conclude the motivational facts by documenting the evolution of labor productivity in Figure 5. The left panel reports sectoral labor-productivity indices, while the right panel reports the corresponding economy-wide aggregate index. Labor productivity grows at an average rate of 2.4% per year between 1950 and 2023 in both Goods and RI-Services. By contrast, labor productivity in NRI-Services grows by only 1.4% per year over the same period. Because all indices are normalized to one in 1950, their 2023 levels summarize the cumulative productivity gains over the sample, while the accompanying average growth rates facilitate comparison across sectors. The aggregate Services sector therefore masks substantial heterogeneity in productivity performance: RI-Services tracks the rapid productivity growth of Goods, while NRI-Services experiences persistently slower growth. The aggregate index in the right panel summarizes the economy-wide productivity outcome associated with these heterogeneous sectoral paths.

Figure 5. Labor Productivity Across Sectors



Notes: The left panel reports Törnqvist labor-productivity indices for Goods, RI-Services, and NRI-Services. The right panel reports the corresponding economy-wide Törnqvist labor-productivity index. All indices are normalized to one in 1950. Legend entries report the 2023 index value and the average annual log growth rate over 1950–2023. The underlying industry-level data are from U.S. KLEMS and the BEA–BLS Integrated Industry-Level Production Accounts; see Subsection 2.1 for details. RI-Services includes Information, Finance and Insurance, and Professional, Scientific and Technical Services; NRI-Services includes all remaining service industries. In the left panel, hollow diamonds, squares, and triangles denote observations for Goods, RI-Services, and NRI-Services, respectively, with marker area proportional to the corresponding sector’s value-added share. In the right panel, hollow hexagons denote observations for the economy-wide index. Dashed lines are cubic smoothing splines with smoothing parameter 10^4 .

2.3 Summary

The empirical evidence reveals a sharp distinction between the sectoral reallocation of production and innovation in the postwar US economy. Innovation activity moves away from Goods toward Services, but almost entirely into RI-Services: both R&D employment and patenting expand substantially in this sector, while their shares in NRI-Services remain broadly flat. Production activity follows a different path. Both Service sectors gain sizable shares of aggregate value added as Goods contracts, and NRI-Services remains by far the largest sector of the economy despite attracting little of the reallocation of innovation activity.

This divergence is mirrored in sectoral productivity. Labor productivity grows at similar rates in Goods and RI-Services, but substantially more slowly in NRI-Services, whose relative price also rises over the postwar period. The sector absorbing the largest share of production activity is therefore not the sector toward which innovation inputs and outputs are being reallocated. These facts characterize the empirical pattern at the heart of Baumol’s cost disease: economic activity shifts toward a service sector with slower productivity growth, while innovative activity is increasingly concentrated elsewhere.

3 Model

This section develops a multi-sector endogenous growth model to study the economic forces governing the joint allocation of production and innovation across sectors. Section 2 documents that production activity shifts toward both service sectors, while innovation activity reallocates primarily toward RI-Services. The model determines these allocation patterns within a common equilibrium and traces their implications for sectoral and aggregate labor productivity. Sectoral productivity differences therefore emerge endogenously as part of the equilibrium allocation.

The model combines the standard forces of structural change with directed technical change. On the production side, a non-homothetic final-good technology generates sectoral reallocation through relative-price and income effects, as in the structural-change literature (Herrendorf et al. 2014). We couple this demand system with a directed technical change block in the tradition of Acemoglu (2002). In an imperfectly competitive environment, innovators develop sector-specific production inputs, and the return to innovation depends on the relative price and market size of the sector in which those inputs are used. Structural change therefore alters not only the allocation of production across sectors, but also the incentives governing the direction of innovation.

Innovation technologies are heterogeneous across sectors. Research productivity evolves differently across research lines, and innovation in each sector draws on knowledge accumulated both within that sector and elsewhere through cross-sector spillovers. Innovators are also heterogeneous by sector in how much markups they charge to the sectoral good technology. The allocation of research employment is therefore shaped by both the value of sector-specific innovation and the productivity of research in each sector. In equilibrium, sectoral labor productivity admits an analytical decomposition into endogenous innovation, capital deepening, and labor efficiency, linking the allocation of production and innovation directly to sectoral and aggregate productivity.

Environment. Time is continuous and indexed by $t \in [0, \infty)$. The economy is populated by an infinitely lived representative household composed of a continuum of identical individuals of measure $L(t)$. Each individual supplies one unit of labor inelastically. Labor is allocated between two occupations, production and research, with the aggregate division between them taken as exogenous. Within each occupation, workers are freely mobile across three sectors: Goods (m), research-intensive Services (rs), and non-research-intensive Services (ns). Population evolves exogenously according to

$$\frac{\dot{L}(t)}{L(t)} = g_\ell(t),$$

with $L(0) = 1$.

The household owns all firms in the economy and has access to perfect capital markets. A competitive final-good producer combines output from Goods (Y_m), research-intensive Services (Y_{rs}), and non-research-intensive Services (Y_{ns}) into a single final good, $Y(t)$, which serves as the numeraire. Within each sector $j \in \{m, rs, ns\}$, a continuum of monopolistically competitive firms produces differentiated varieties ω by combining production labor with sector-specific bundles of machines.

Machines are supplied by monopolistically competitive intermediate producers. Innovation is carried out by research labs and takes the form of creating new machine varieties. Labs hire scientists to develop sector-specific machine blueprints, and a successful innovation grants a patent conferring the exclusive right to produce the corresponding machine variety.

The stream of profits generated by a machine variety determines the value of its patent and thereby contributes to the return to research directed toward that sector. The allocation of research across sectors is therefore endogenous and responds to sectoral innovation incentives.

Aggregate Production and Demand Structure. The competitive final-good producer combines sectoral outputs using the non-homothetic CES aggregator of [Hanoch \(1975\)](#), parameterized as in [Comin et al. \(2021\)](#). The aggregator is defined implicitly by

$$\sum_{j \in \{m, rs, ns\}} \theta_j^{1/\sigma} \left(\frac{Y_j(t)}{Y(t)^{\eta_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1, \quad (1)$$

where θ_j are sectoral weights and σ governs the elasticity of substitution across sectoral outputs. The Engel parameters η_j govern non-homotheticity and therefore the response of sectoral demand to aggregate income. For $\sigma < 1$, as in our calibration, sectors with higher η_j have higher income elasticities of demand, generating income-driven reallocation across sectors as aggregate output grows.¹²

Sectoral Production Technologies. In each sector $j \in \{m, rs, ns\}$, a continuum of monopolistically competitive firms produces differentiated varieties of sectoral output, indexed by $\omega \in [0, 1]$. A competitive sectoral producer combines these varieties into

¹²Holding sectoral prices fixed, the income elasticity of demand for sector j implied by Equation (1) is

$$\frac{\partial \ln Y_j(t)}{\partial \ln Y(t)} = \sigma + (1 - \sigma)\eta_j.$$

The homothetic case corresponds to $\eta_j = 1$ for all j .

sectoral output $Y_j(t)$ according to

$$Y_j(t) = \left(\int_0^1 y_j(\omega, t)^{\mu_j^y(t)} d\omega \right)^{1/\mu_j^y(t)}, \quad \mu_j^y(t) \in (0, 1), \quad (2)$$

where $y_j(\omega, t)$ denotes output of variety ω in sector j . The elasticity of substitution across varieties is $1/(1 - \mu_j^y(t))$, implying an equilibrium markup of $1/\mu_j^y(t)$ over marginal cost. A higher $\mu_j^y(t)$ therefore corresponds to a lower variety-level markup.

Each variety producer combines production labor with a sector-specific bundle of machines. The technology of a firm producing variety ω in sector j is

$$y_j(\omega, t) = X_j(\omega, t)^{1-\alpha} (\bar{h}_j(t) L_j^p(\omega, t))^\alpha, \quad (3)$$

where $L_j^p(\omega, t)$ denotes production labor, $\bar{h}_j(t)$ is an exogenous labor-augmenting productivity term common to all firms in sector j , $X_j(\omega, t)$ is the machine composite, and $\alpha \in (0, 1)$ is the elasticity of output with respect to production labor, common across sectors and time. Under variety-level price setting, production labor receives the share $\alpha\mu_j^y(t)$ of sectoral revenue, with the wedge relative to α reflecting the variety-level markup.

The machine composite is constructed from a continuum of sector-specific machine varieties indexed by $\nu \in [0, N_j(t)]$:

$$X_j(\omega, t) = N_j(t)^{\kappa_j} \left(\int_0^{N_j(t)} x_j(\omega, \nu, t)^{\mu_j^x(t)} d\nu \right)^{1/\mu_j^x(t)}, \quad \mu_j^x(t) \in (0, 1), \quad \kappa_j > 0, \quad (4)$$

where $x_j(\omega, \nu, t)$ denotes the quantity of machine variety ν used by the firm. The scaling term $N_j(t)^{\kappa_j}$ introduces an additional love-of-variety effect beyond that implied by the CES aggregator, following [Benassy \(1996\)](#). In a standard [Dixit & Stiglitz \(1977\)](#) aggregator, the strength of the love-of-variety effect is tied to the elasticity of substitution across machine varieties. The term $N_j(t)^{\kappa_j}$ separates these two forces, allowing the strength of the additional variety effect to differ across sectors independently of $\mu_j^x(t)$. We treat κ_j as constant over time but allow it to vary across sectors.

This nested production structure generates two layers of imperfect competition within each sector, one among producers of sectoral output varieties and another among machine producers. An expansion in the set of available machine varieties $N_j(t)$ raises sectoral productivity through love-of-variety, with the strength of this effect governed jointly by $\mu_j^x(t)$ and κ_j .

Machine Producers. In each sector $j \in \{m, rs, ns\}$, a continuum of monopolistically competitive firms produces sector-specific machine varieties indexed by ν . Each firm

produces a single variety, which can be used only by firms in sector j .

Machine producers transform the economy-wide final good into machines using a linear technology. Let $y_j^x(\nu, t)$ denote the quantity of the final good used to produce machine variety ν destined for sector j .¹³ Machine output satisfies

$$x_j(\nu, t) = z_j(t)y_j^x(\nu, t),$$

where $z_j(t)$ is a sector-specific productivity parameter common to all machine producers supplying sector j . Hence, producing one unit of a machine for sector j requires $1/z_j(t)$ units of the final good.

Research and Development. Innovation is carried out by a continuum of research labs that can freely enter each sectoral research line. Labs hire scientists to develop machine blueprints specific to a given sector, and each successful innovation expands the set of machine varieties available in that sector, $N_j(t)$.

When a research lab discovers a new blueprint for sector j , it obtains a fully enforced patent granting the exclusive right to produce the corresponding machine variety. Blueprints do not depreciate and therefore permanently expand the stock of available machine varieties. Machines produced from these blueprints, by contrast, are fully used up in production at each instant and must be reproduced continuously.

The stock of machine varieties in sector j evolves according to

$$\dot{N}_j(t) = \zeta_j(t) \Lambda_j(t)^\gamma L_j^s(t)^\phi, \quad \Lambda_j(t) = \prod_{i \in \{m, rs, ns\}} \tilde{N}_i(t)^{\delta_{ji}(t)}, \quad \tilde{N}_i(t) \equiv \bar{N}_i N_i(t), \quad (5)$$

where $L_j^s(t)$ denotes the number of scientists employed in research directed toward sector j , and $\zeta_j(t)$ is a sector-specific, time-varying research-productivity term. Variation in $\zeta_j(t)$ captures shifts in the productivity of research effort that are not accounted for by accumulated knowledge or the scale of research employment. The parameters $\gamma \in (0, 1)$ and $\phi \in (0, 1)$ govern the elasticities of new idea creation with respect to the effective knowledge stock and research labor, respectively.

The term $\Lambda_j(t)$ aggregates knowledge accumulated across sectors and captures cross-sector knowledge spillovers. The weight $\delta_{ji}(t) \in [0, 1]$ is the elasticity of the effective knowledge stock in research line j with respect to knowledge originating in sector i . We impose

$$\sum_{i \in \{m, rs, ns\}} \delta_{ji}(t) = 1$$

¹³Notice that the subscript j indexes the sector in which the machine is used, not the origin of the input used in its production.

for every sector j and date t , so that a proportional increase in all knowledge stocks raises $\Lambda_j(t)$ in the same proportion. The constants $\bar{N}_i > 0$ allow inherited knowledge stocks to differ across sectors, while $N_i(t)$ captures their subsequent endogenous expansion.

Market Value of Patents. Research labs sell newly discovered machine blueprints, together with the associated patents, to machine producers. A patent grants its owner the exclusive right to produce the corresponding machine variety and sell it to firms in sector j .

Let

$$x_j(\nu, t) \equiv \int_0^1 x_j(\omega, \nu, t) d\omega$$

denote aggregate demand for machine variety ν from firms in sector j , and let $p_j^x(\nu, t)$ denote its price. The instantaneous operating profit of the machine producer is

$$\pi_j^x(\nu, t) = p_j^x(\nu, t)x_j(\nu, t) - y_j^x(\nu, t), \quad j \in \{m, rs, ns\},$$

where $y_j^x(\nu, t)$ is the quantity of the economy-wide final good used to produce the machine.

Let $V_j(\nu, t)$ denote the ex-dividend market value of the patent. Since patents do not expire, their value equals the present discounted value of future operating profits,

$$V_j(\nu, t) = \int_t^\infty e^{-\int_t^s r(s') ds'} \pi_j^x(\nu, s) ds, \quad j \in \{m, rs, ns\}.$$

Equivalently, no arbitrage requires

$$r(t)V_j(\nu, t) = \pi_j^x(\nu, t) + \dot{V}_j(\nu, t), \quad j \in \{m, rs, ns\},$$

subject to the corresponding transversality condition.

Allocation of Labor. The labor force is divided between production and research. Let

$$\ell^s(t) \equiv \frac{L^s(t)}{L(t)}, \quad \ell^p(t) \equiv \frac{L^p(t)}{L(t)} = 1 - \ell^s(t)$$

denote the shares of the labor force employed in research and production, respectively. The division between the two occupations is exogenous, so

$$L^s(t) = \ell^s(t)L(t), \quad L^p(t) = (1 - \ell^s(t))L(t). \quad (6)$$

Within each occupation, workers move freely across sectors. Production wages therefore

equalize,

$$w^p(t) \equiv w_m^p(t) = w_{rs}^p(t) = w_{ns}^p(t),$$

and research wages satisfy

$$w^s(t) \equiv w_m^s(t) = w_{rs}^s(t) = w_{ns}^s(t).$$

Taking $\ell^s(t)$ as given conditions the model on the aggregate scale of research employment, while its allocation across research lines is determined endogenously by innovation incentives.¹⁴

Representative Household. The representative household has CRRA preferences over per-capita consumption of the final good. Let $C(t)$ denote aggregate consumption and $c(t) \equiv C(t)/L(t)$ per-capita consumption. Lifetime utility is

$$\int_0^\infty e^{-\rho t} L(t) \frac{c(t)^{1-\chi} - 1}{1-\chi} dt,$$

where $\rho > 0$ is the subjective discount rate and $\chi > 0$ is the inverse of the intertemporal elasticity of substitution.¹⁵

The household owns all firms in the economy and trades ownership claims in perfect capital markets. Let $B(t)$ denote the market value of household financial wealth and $b(t) \equiv B(t)/L(t)$ financial wealth per capita. No arbitrage implies that all ownership claims earn the common instantaneous total return $r(t)$. Using the occupational shares defined above, the household's per-capita budget constraint is

$$\dot{c}(t) + \dot{b}(t) = w^p(t)\ell^p(t) + w^s(t)\ell^s(t) + (r(t) - g_\ell(t))b(t).$$

Firm profits do not enter separately because dividends and capital gains are incorporated in the total return on household financial wealth. The household is endowed with initial financial wealth $b(0) > 0$.

Maximization of lifetime utility subject to the budget constraint yields the standard Euler equation

$$\frac{\dot{c}(t)}{c(t)} = \frac{r(t) - \rho}{\chi},$$

¹⁴Differences between production and research wages may reflect differences in training and human-capital requirements across occupations or occupational mobility frictions, which the model does not endogenize.

¹⁵Lifetime utility is finite under the usual restrictions on population and consumption growth. Along a balanced growth path with constant population growth g_ℓ and per-capita consumption growth g_c , sufficient conditions are $\rho > g_\ell$ and $\rho > g_\ell + (1 - \chi)g_c$. When $\chi \geq 1$ and $g_c \geq 0$, the first condition implies the second.

which governs the intertemporal allocation of consumption. We impose the transversality condition

$$\lim_{t \rightarrow \infty} e^{-\int_0^t r(s) ds} L(t) b(t) = 0.$$

3.1 Equilibrium

We now characterize the competitive equilibrium of the economy. Throughout, the price of the final good is normalized to one. The derivations of the equilibrium conditions stated below are collected in Appendix J.

Definition 3.1 (Competitive equilibrium). *Given exogenous paths*

$$\{g_\ell(t), \ell^s(t), \bar{h}_j(t), z_j(t), \mu_j^y(t), \mu_j^x(t), \zeta_j(t), \delta_{ji}(t)\},$$

parameters

$$\{\theta_j, \eta_j, \sigma, \alpha, \kappa_j, \phi, \gamma, \bar{N}_j, \rho, \chi\},$$

and initial conditions $\{N_j(0)\}$ and $b(0)$, with $L(0) = 1$, a competitive equilibrium is a collection of allocations

$$\{C(t), b(t), Y(t), Y_j(t), y_j(\omega, t), L_j^p(\omega, t), L_j^s(t), \\ X_j(\omega, t), x_j(\omega, \nu, t), y_j^x(\nu, t), N_j(t)\}$$

and prices

$$\{p_j(t), p_j(\omega, t), p_j^X(t), p_j^x(\nu, t), w_j^p(t), w_j^s(t), V_j(\nu, t), r(t)\}$$

such that: (i) the representative household optimizes; (ii) the final-good producer, sectoral aggregators, variety producers, machine producers, and research labs solve their respective optimization problems; (iii) free entry into innovation holds in each research line; (iv) production and research labor markets clear, given the exogenous research share $\ell^s(t)$; (v) final-good and machine markets clear; and (vi) markets for ownership claims clear.

Population $L(t)$ is determined by $g_\ell(t)$ and the normalization $L(0) = 1$, while aggregate production and research employment satisfy

$$L^p(t) + L^s(t) = L(t).$$

We characterize the equilibrium by proceeding through the production structure of the economy. The final-good producer determines sectoral expenditure, variety producers demand labor and machines, machine producers set prices and generate the profit flows that determine patent values, and free entry and worker mobility determine the allocation

of research and production labor across sectors. Market-clearing conditions then close the equilibrium.

Sectoral Demands and Price Index. The final-good producer chooses $\{Y_m(t), Y_{rs}(t), Y_{ns}(t)\}$ to minimize the cost of producing $Y(t)$ using the technology in Equation (1), taking sectoral prices $\{p_m(t), p_{rs}(t), p_{ns}(t)\}$ as given. Cost minimization yields

$$Y_j(t) = \frac{s_j(t)Y(t)}{p_j(t)},$$

where

$$s_j(t) \equiv \theta_j p_j(t)^{1-\sigma} Y(t)^{(\eta_j-1)(1-\sigma)}, \quad j \in \{m, rs, ns\}. \quad (7)$$

With the price of the final good normalized to one, the corresponding price-index condition implies

$$\sum_j s_j(t) = 1.$$

Hence,

$$s_j(t) = \frac{p_j(t)Y_j(t)}{Y(t)}$$

is the expenditure share of the final-good producer on sector j , equivalently the sector's share of gross revenue. These expenditure shares need not coincide with sectoral value-added shares because machine production uses the final good as an intermediate input. We characterize the mapping between the two below when imposing final-good market clearing.

Variety Producers and Machine Demand. Within each sector j , a competitive sectoral producer combines differentiated varieties into $Y_j(t)$ using the technology in Equation (2). Taking variety prices as given, cost minimization yields the demand schedule

$$y_j(\omega, t) = \left(\frac{p_j(\omega, t)}{p_j(t)} \right)^{-\frac{1}{1-\mu_j^y(t)}} Y_j(t),$$

and the sectoral price index

$$p_j(t) = \left(\int_0^1 p_j(\omega, t)^{\frac{\mu_j^y(t)}{\mu_j^y(t)-1}} d\omega \right)^{\frac{\mu_j^y(t)-1}{\mu_j^y(t)}}.$$

Each variety producer takes this demand schedule, the production wage $w_j^p(t)$, and the price of the machine composite $p_j^X(t)$ as given and maximizes instantaneous profits. Marginal revenue is $\mu_j^y(t)$ times the variety price, implying a markup $1/\mu_j^y(t)$ over

the Cobb–Douglas marginal cost associated with Equation (3). Cost minimization over machine varieties yields the inverse demand

$$p_j^x(\nu, t) = p_j^X(t) N_j(t)^{\kappa_j \mu_j^x(t)} \left(\frac{X_j(t)}{x_j(\nu, t)} \right)^{1-\mu_j^x(t)}.$$

The term $N_j(t)^{\kappa_j \mu_j^x(t)}$ reflects the additional love-of-variety component introduced in Equation (4).¹⁶

All varieties within a sector face the same technology and factor prices. The symmetric equilibrium therefore satisfies

$$p_j(\omega, t) = p_j(t), \quad y_j(\omega, t) = Y_j(t), \quad L_j^p(\omega, t) = L_j^p(t), \quad X_j(\omega, t) = X_j(t).$$

The first-order conditions then imply

$$w_j^p(t) = \mu_j^y(t) \alpha \frac{p_j(t) Y_j(t)}{L_j^p(t)}, \quad (8)$$

and

$$p_j^X(t) = \mu_j^y(t) (1 - \alpha) \frac{p_j(t) Y_j(t)}{X_j(t)}.$$

Thus, production labor receives the share $\alpha \mu_j^y(t)$ of sectoral revenue, while expenditure on the machine composite accounts for the share $(1 - \alpha) \mu_j^y(t)$. The remaining share $1 - \mu_j^y(t)$ accrues to variety producers as operating profits.

Machine Supply and Pricing. A producer of machine variety ν for sector j has technology

$$x_j(\nu, t) = z_j(t) y_j^x(\nu, t)$$

and faces the inverse demand derived above. Profit maximization yields

$$p_j^x(\nu, t) = \frac{1}{\mu_j^x(t)} \frac{1}{z_j(t)}, \quad (9)$$

so the machine producer charges a markup $1/\mu_j^x(t)$ over marginal cost $1/z_j(t)$.

The machine composite is homogeneous of degree one in machine quantities. Combining total expenditure on machines, $p_j^X(t) X_j(t)$, with Equation (9) yields the symmetric per-variety quantity

$$x_j(\nu, t) = x_j(t) = \mu_j^y(t) \mu_j^x(t) (1 - \alpha) z_j(t) \frac{p_j(t) Y_j(t)}{N_j(t)}.$$

¹⁶See Appendix ?? for a formal derivation of these optimality conditions.

Per-variety operating profits are therefore

$$\pi_j^x(\nu, t) = \frac{x_j(t)}{z_j(t)} \frac{1 - \mu_j^x(t)}{\mu_j^x(t)}.$$

A rise in $z_j(t)$ lowers the per-unit margin but raises the quantity sold in the same proportion, conditional on sectoral revenue. These effects cancel, leaving profits proportional to sectoral revenue per machine variety:

$$\pi_j^x(\nu, t) = \mu_j^y(t)(1 - \mu_j^x(t))(1 - \alpha) \frac{p_j(t)Y_j(t)}{N_j(t)}. \quad (10)$$

Machine productivity $z_j(t)$ therefore does not enter patent profits directly. It affects the return to innovation through its general-equilibrium effect on sectoral prices and expenditure. A productivity improvement in sector j lowers its relative price; for $\sigma < 1$, holding aggregate output fixed, this reduces the sector's expenditure share and hence the profit flow associated with a new machine variety. Productivity growth can therefore alter the direction of innovation through the same relative-price forces that drive structural change.

Sectoral Labor Productivity. The equilibrium conditions for variety producers and machine suppliers jointly determine the amount and composition of machine inputs used in each sector. Combining these conditions yields a closed-form expression for sectoral labor productivity, which will play a central role in the quantitative analysis.

Proposition 1 (Sectoral Labor Productivity). *In the symmetric equilibrium, sectoral labor productivity is*

$$\mathcal{A}_j(t) \equiv \frac{Y_j(t)}{L_j^p(t)} = N_j(t)e_j^Y(t) [\mu_j^y(t)\mu_j^x(t)(1 - \alpha)p_j(t)z_j(t)]^{\left(\frac{1-\alpha}{\alpha}\right)} \bar{h}_j(t), \quad (11)$$

where

$$e_j^Y(t) \equiv \left(\kappa_j + \frac{1 - \mu_j^x(t)}{\mu_j^x(t)} \right) \left(\frac{1 - \alpha}{\alpha} \right).$$

Proof. See Appendix J.1. □

Proposition 1 shows that sectoral labor productivity can be decomposed as

$$\mathcal{A}_j(t) = \underbrace{N_j(t)e_j^Y(t)}_{\text{Endogenous innovation}} \cdot \underbrace{[\mu_j^y(t)\mu_j^x(t)(1 - \alpha)p_j(t)z_j(t)]^{\left(\frac{1-\alpha}{\alpha}\right)}}_{\text{Capital deepening}} \cdot \underbrace{\bar{h}_j(t)}_{\text{Labor efficiency}}.$$

The first component captures productivity gains from the endogenous expansion of ma-

chine varieties. Its strength depends on both the standard love-of-variety effect generated by the CES machine aggregator and the additional variety effect governed by κ_j .

The second component captures capital deepening. Equation (9) implies

$$\frac{p_j(t)}{p_j^x(t)} = \mu_j^x(t) p_j(t) z_j(t),$$

so optimal machine use per worker rises with the price of sectoral output relative to the price of machines, while $\mu_j^y(t)$ governs the share of sectoral revenue allocated to machine expenditure. The final component captures exogenous sector-specific labor efficiency.

Combining Proposition 1 with Equation (8), the production wage can be written as

$$w_j^p(t) = \alpha \mu_j^y(t) p_j(t) \mathcal{A}_j(t),$$

which is the reduced-form expression used below.

Free Entry Into Innovation. Research is organized into sector-specific research lines. Free entry implies that the market value of the flow of newly created patents in sector j is exhausted by the wage bill of scientists employed in that research line:

$$\dot{N}_j(t) V_j(t) = w_j^s(t) L_j^s(t).$$

Using the research technology in Equation (5), the research wage is therefore

$$w_j^s(t) = \zeta_j(t) \Lambda_j(t)^\gamma L_j^s(t)^{\phi-1} V_j(t). \quad (12)$$

Thus, the research wage equals the average value product of research labor, $\dot{N}_j(t)/L_j^s(t)$, valued at the market price of a newly created patent.

Labor Mobility Across Sectors. Production workers move freely across sectors, so production wages are equalized at $w^p(t)$. Using the reduced-form wage derived above, wage equalization implies

$$\frac{p_k(t)}{p_m(t)} = \frac{\mu_m^y(t)}{\mu_k^y(t)} \left[\frac{N_m(t) e_m^Y(t) \bar{h}_m(t)}{N_k(t) e_k^Y(t) \bar{h}_k(t)} \right]^\alpha \left[\frac{\mu_m^x(t) z_m(t)}{\mu_k^x(t) z_k(t)} \right]^{1-\alpha}, \quad k \in \{rs, ns\}. \quad (13)$$

Relative prices therefore fall in sectors with faster machine-variety expansion, stronger labor efficiency, or higher machine productivity, controlling for sectoral markup wedges. When markups and the strength of the love-of-variety channel are common across sectors, this relationship reduces to the standard inverse association between relative productivity

and relative prices familiar from models of structural change.¹⁷ Equation (13) captures the relative-price force underlying Baumol's cost disease: faster productivity growth in a sector lowers its relative price, which, for $\sigma < 1$, reduces its expenditure share through the relative-price channel.

Scientists are likewise freely mobile across research lines, so

$$w_m^s(t) = w_{rs}^s(t) = w_{ns}^s(t).$$

Combining wage equalization with Equation (12) and research labor market clearing, $\sum_j L_j^s(t) = L^s(t)$, yields

$$\frac{L_j^s(t)}{L^s(t)} = \frac{[V_j(t)\zeta_j(t)\Lambda_j(t)^\gamma]^{\frac{1}{1-\phi}}}{\sum_{i \in \{m, rs, ns\}} [V_i(t)\zeta_i(t)\Lambda_i(t)^\gamma]^{\frac{1}{1-\phi}}}, \quad j \in \{m, rs, ns\}, \quad (14)$$

where aggregate research employment $L^s(t)$ is given by Equation (6). Research effort is directed toward sectors with higher patent values, higher research productivity, and larger effective knowledge stocks. The parameter ϕ governs the responsiveness of research employment to these differences: a larger ϕ implies weaker diminishing returns to research labor and therefore a more elastic reallocation of scientists across research lines. Equation (14) characterizes the direction of technical change in the model.

Market Clearing. Sectoral expenditure determines sectoral revenue according to

$$p_j(t)Y_j(t) = s_j(t)Y(t).$$

Using Equation (8), conditional production-labor demand is

$$L_j^p(t) = s_j(t)\mu_j^y(t)\alpha \frac{Y(t)}{w^p(t)}.$$

Summing across sectors and imposing production-labor market clearing,

$$\sum_j L_j^p(t) = L^p(t),$$

gives

$$w^p(t) = \alpha \bar{\mu}^y(t) \frac{Y(t)}{L^p(t)}, \quad \bar{\mu}^y(t) \equiv \sum_j s_j(t)\mu_j^y(t),$$

¹⁷See, for example, [Herrendorf et al. \(2014\)](#).

where $\bar{\mu}^y(t)$ is the expenditure-share-weighted average variety-level markup wedge. Substituting back yields

$$\frac{L_j^p(t)}{L^p(t)} = s_j(t) \frac{\mu_j^y(t)}{\bar{\mu}^y(t)}, \quad Y(t) = \frac{\mu_j^y(t)}{\bar{\mu}^y(t)} p_j(t) \mathcal{A}_j(t) L^p(t), \quad j \in \{m, rs, ns\}. \quad (15)$$

When variety-level markups are common across sectors, production-labor shares coincide with expenditure shares. With markup dispersion, the two differ: conditional on expenditure shares, sectors with higher markups, and hence lower $\mu_j^y(t)$, employ a smaller share of production labor. Notice that machine markups do not enter Equation (15) directly, but affect labor allocation indirectly through sectoral productivity, relative prices, and expenditure shares.

Final-Good Market Clearing. Machine production uses the final good as its only input. Since machines are fully used up within each instant, final-good use by machine producers supplying sector j is

$$I_j^x(t) = \frac{1}{z_j(t)} \int_0^{N_j(t)} x_j(\nu, t) d\nu.$$

Using the symmetric machine demand derived above,

$$I_j^x(t) = \mu_j^y(t) \mu_j^x(t) (1 - \alpha) p_j(t) Y_j(t).$$

Aggregating across sectors gives

$$I^x(t) = (1 - \alpha) \overline{\mu^y \mu^x}(t) Y(t), \quad \overline{\mu^y \mu^x}(t) \equiv \sum_j s_j(t) \mu_j^y(t) \mu_j^x(t),$$

where $\overline{\mu^y \mu^x}(t)$ is the expenditure-share-weighted machine-input wedge.

Final-good market clearing requires

$$Y(t) = C(t) + I^x(t),$$

and therefore

$$C(t) = [1 - (1 - \alpha) \overline{\mu^y \mu^x}(t)] Y(t). \quad (16)$$

The same resource accounting provides the mapping from expenditure shares to sectoral value-added shares. Define

$$\omega_j(t) \equiv 1 - (1 - \alpha) \mu_j^y(t) \mu_j^x(t).$$

Consolidating sectoral producers with the machine producers that supply them, nominal value added in sector j is

$$VA_j(t) = p_j(t)Y_j(t) - I_j^x(t) = \omega_j(t)p_j(t)Y_j(t).$$

Hence, sector j 's share of aggregate value added is

$$s_j^{VA}(t) = \frac{\omega_j(t)s_j(t)}{\sum_i \omega_i(t)s_i(t)}. \quad (17)$$

Expenditure shares and value-added shares therefore coincide when the markup-driven wedges $\omega_j(t)$ are common across sectors. Finally, the household Euler equation implies

$$r(t) = \rho + \chi \frac{\dot{c}(t)}{c(t)}.$$

Using Equation (16),

$$\frac{\dot{c}(t)}{c(t)} = \frac{\dot{Y}(t)}{Y(t)} - g_\ell(t) + \frac{d}{dt} \ln[1 - (1 - \alpha)\overline{\mu^y \mu^x}(t)].$$

Thus, per-capita consumption growth equals per-capita final-output growth plus the change in the share of final output available for consumption after accounting for machine production.

Model Summary. The equilibrium jointly determines the allocation of production and innovation across sectors. Relative-price and income effects govern sectoral expenditure, while worker mobility maps expenditure into the allocation of production labor. At the same time, scientists are directed toward research lines offering higher patent values, higher research productivity, and larger effective knowledge stocks. Because patent values depend on the profitability of sector-specific machine varieties, structural change and markups feed back into the direction of innovation.

These interactions determine sectoral productivity endogenously. Proposition 1 shows that labor productivity depends on the accumulation of machine varieties, capital deepening, and labor efficiency. The allocation of production therefore shapes the incentives to innovate, while the resulting innovation changes sectoral productivity and relative prices and feeds back into structural change. The calibration disciplines these forces using the joint evolution of production, innovation, and productivity documented in Section 2.

4 Calibration

We now take the model to the data, combining externally disciplined parameters and directly measured paths with parameters calibrated internally from the joint evolution of production, innovation, and productivity.

The time-invariant parameters govern household preferences, innovation technology, and sectoral production and demand. They include the preference parameters $\{\rho, \chi\}$, the innovation elasticities $\{\gamma, \phi\}$, the initial knowledge stocks $\{\bar{N}_i\}$ and the production and demand parameters $\{\sigma, \alpha, \eta_j, \theta_j, \kappa_j\}$. The time-varying objects include sectoral output and machine markup wedges $\{\mu_j^y(t), \mu_j^x(t)\}$, research productivity $\zeta_j(t)$, labor efficiency $\bar{h}_j(t)$, machine productivity $z_j(t)$, and the cross-sector knowledge-spillover elasticities $\delta_{ji}(t)$, for $j, i \in \{m, rs, ns\}$. We also take the aggregate research-employment share $\ell^s(t)$ and population growth $g_\ell(t)$ as exogenous paths. We discipline these objects using a combination of external calibration, direct measurement, and internal calibration.

4.1 External Calibration

We externally discipline a subset of the time-invariant parameters and most of the time-varying objects before the internal calibration. The externally set time-invariant parameters are the household preference parameters $\{\rho, \chi\}$, the innovation elasticities $\{\gamma, \phi\}$, the initial knowledge stocks $\{\bar{N}_i\}$ and the labor elasticity in sectoral production, α . These are summarized in Table 1. Among the time-varying objects, all paths other than sectoral research productivity $\zeta_j(t)$ are disciplined directly from the data or external estimates, as described below and summarized in Figure 7.

4.1.1 Time-Invariant Parameters

Preferences. The household preference parameters are the subjective discount rate, ρ , and the inverse intertemporal elasticity of substitution, χ . We set $\rho = 0.04$ and $\chi = 2$, implying an intertemporal elasticity of substitution of $1/\chi = 0.5$. These values are standard in the quantitative macroeconomic literature (Summers 1981, Huggett 1993).

Innovation Technology. We set both the research-labor elasticity, ϕ , and the aggregate knowledge-spillover elasticity, γ , equal to 0.50, in line with the benchmark values in Jones (1995).

Initial Knowledge Stocks. We normalize the production-side variety stocks to $N_j(1950) = 1$ for all sectors. To preserve differences in inherited knowledge across sectors, we set the constants $\bar{N}_i$ equal to each sector's share of the patent stock in 1950. Since $N_i(1950) = 1$, these values determine the initial effective knowledge stocks $\tilde{N}_i(1950) = \bar{N}_i$ entering

the spillover aggregator, while their subsequent evolution is driven by endogenous idea accumulation. We obtain values of 0.97, 0.01 and 0.02 for $\bar{N}_m$, $\bar{N}_{rs}$ and $\bar{N}_{ns}$, respectively.

Sectoral Production. We set the labor elasticity in sectoral production to $\alpha = 0.70$, consistent with the benchmark evidence on labor’s share of income in [Gollin \(2002\)](#). We treat α as externally set and hold it fixed throughout the calibration.

Table 1. Externally Set Time-Invariant Parameters

Parameter	Interpretation	Value	Source
ρ	Subjective discount rate	0.04	(Huggett 1993)
χ	Inverse intertemporal elasticity of substitution	2.00	(Summers 1981)
γ	Aggregate knowledge-spillover elasticity	0.50	(Jones 1995)
ϕ	Research-labor elasticity	0.50	(Jones 1995)
$\bar{N}_m$	Goods initial knowledge stock	0.97	(Marco et al. 2015)
$\bar{N}_{rs}$	RI-Services initial knowledge stock	0.01	(Marco et al. 2015)
$\bar{N}_{ns}$	NRI-Services initial knowledge stock	0.02	(Marco et al. 2015)
α	Labor elasticity in sectoral production	0.70	(Gollin 2002)

Notes: The table reports time-invariant parameters set prior to the internal calibration. The value of α is chosen to be consistent with benchmark evidence on labor’s share of income.

4.1.2 Time-Varying Parameters

Sectoral Variety Markups. We infer the sectoral output markup wedges $\mu_j^y(t)$ from observed labor-compensation shares of value added. Using the equilibrium conditions derived in Section 3,

$$\frac{w_j^p(t)L_j^p(t)}{VA_j(t)} = \frac{\alpha\mu_j^y(t)}{1 - (1 - \alpha)\mu_j^y(t)\mu_j^x(t)}.$$

Given α and the externally disciplined machine markup wedge $\mu_j^x(t)$, this relationship identifies $\mu_j^y(t)$ from the sectoral labor-compensation share. We construct these shares using the data in [Elbadri & Vizcaino \(2025\)](#).

The top-left panel of Figure 7 shows substantial heterogeneity in the evolution of sectoral market power. Output markups rise markedly in both Goods and NRI-Services, while the increase is more limited in RI-Services. Because Goods and NRI-Services together account for a large share of aggregate value added, these trends imply that rising market power is not confined to a narrow part of the economy. This pattern is consistent with the broader evidence of rising markups in the United States since the 1980s ([?De Loecker et al. 2020, ?](#)).

Machine Producer Markups. We discipline machine-producer markups using the decomposition of sectoral capital compensation into the user cost of capital and profits. We construct user costs from data on capital prices, depreciation rates, capital gains, and

capital stocks, following the BLS Handbook of Methods ([U.S. Bureau of Labor Statistics 2025a](#)) and the user-cost approach of [Jorgenson \(1963\)](#). The residual component of capital compensation is attributed to profits.

In the model, the machine-producer markup is $1/\mu_j^x(t)$, and the corresponding profit share of machine revenue is $1 - \mu_j^x(t)$. We therefore use the empirical profit share of capital compensation to discipline the sector-specific path of $\mu_j^x(t)$.¹⁸

The top-right panel of Figure 7 reports the implied machine-producer markups. RI-Services exhibits higher machine-producer market power than Goods and NRI-Services through much of the second half of the twentieth century. By the end of the sample, however, machine-producer markups converge toward similar and lower levels across sectors.

Machine Producer Productivity. Equation (9) implies that the price of a machine, relative to the final good, satisfies

$$p_j^x(t) = \frac{1}{\mu_j^x(t)z_j(t)}.$$

We construct sector-level relative machine-price indices from the same asset-by-industry data used to measure machine-producer markups, following the aggregation procedure described in Appendix G. Given the path of $\mu_j^x(t)$, changes in relative machine prices identify changes in machine productivity through Equation (9). We normalize $z_j(1950) = 1$ for each sector and assume that machine productivity thereafter grows at a constant rate equal to the average annual growth rate implied by the data over 1950–2023.

The middle-left panel of Figure 7 shows substantially faster machine-productivity growth in RI-Services than in Goods or NRI-Services. This pattern reflects the greater importance in RI-Services of software and information-technology equipment, whose relative prices have fallen sharply over the sample period ([Greenwood et al. 1997](#), [Caunedo & Keller 2024](#)).

Labor-Augmenting Productivity. We discipline labor-augmenting productivity, $\bar{h}_j(t)$, using harmonized industry-level data from KLEMS and the BEA, as described in Section 2. We use labor Services indices and hours worked to measure labor quality growth as the difference between growth in labor Services and growth in hours. We then aggregate industry-level labor quality growth to our three sectors using Törnqvist value-added weights. We normalize $\bar{h}_j(1950) = 1$ and assume constant growth at the sector-specific

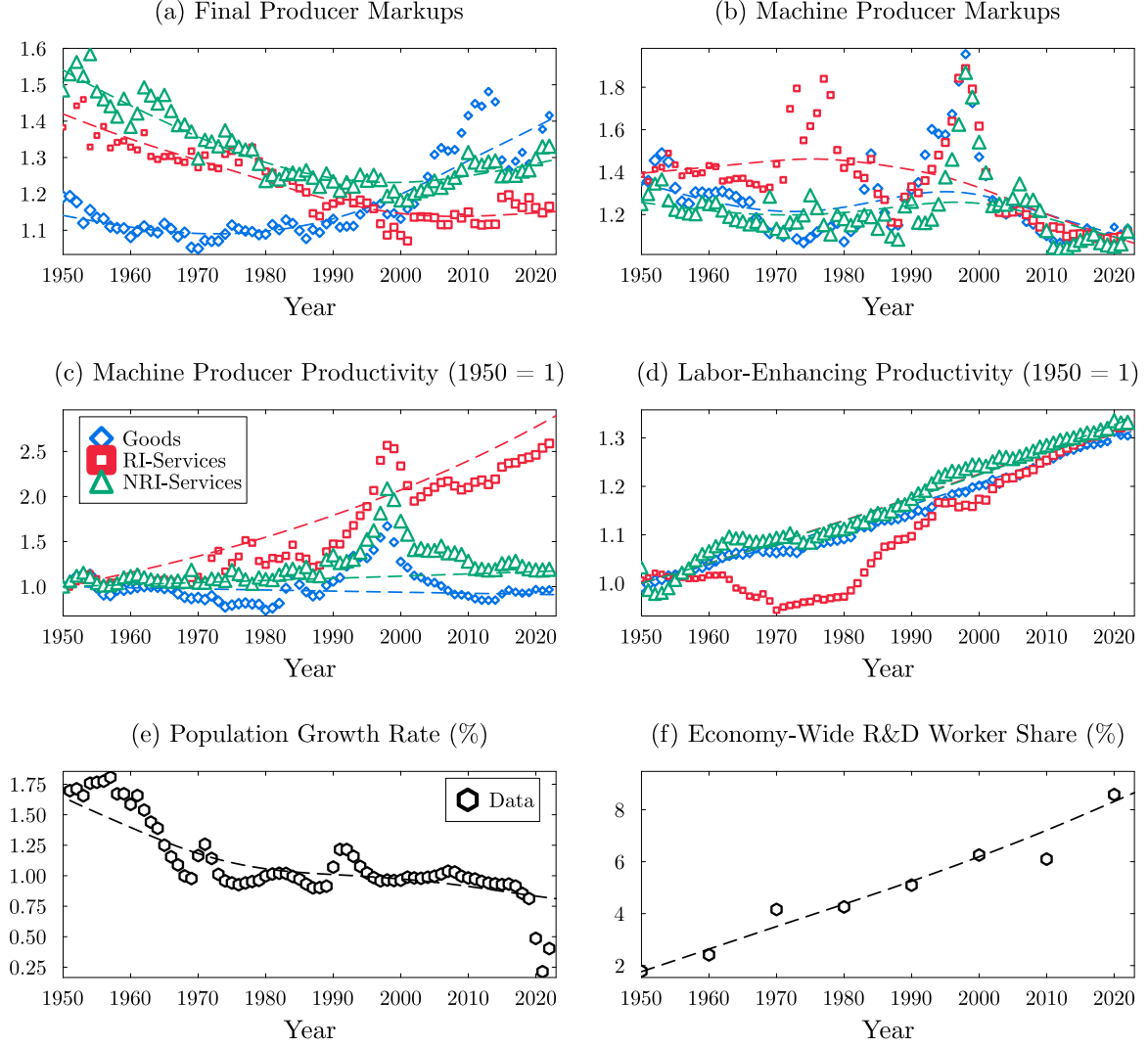
¹⁸The BLS distinguishes between an external rate of return, used as a benchmark opportunity cost of capital, and an internal rate of return that exhausts observed capital compensation given the capital stock. We use the resulting difference in capital income to separate user costs from profits. Appendix F describes the construction and data sources in detail.

average rate over 1950–2023. The middle-right panel of Figure 7 shows similar labor-augmenting productivity growth across the three sectors.

Population. We obtain U.S. population data from Penn World Table 11.0 (Feenstra et al. 2015). We normalize $L(1950) = 1$ and construct the population path using a smoothed time-varying growth rate, $g_\ell(t)$, estimated from the data. The bottom-left panel of Figure 7 reports the resulting population growth rate over 1950–2023.

Economy-Wide R&D Employment Shares. We construct the economy-wide scientist share, $\ell^s(t)$, using the IPUMS data underlying our measures of structural change in R&D employment in Figures 1 and 2. We identify R&D workers at the individual level, aggregate them across industries, and divide by total employment. Appendix B describes the construction in detail. The bottom-right panel of Figure 7 shows that the scientist share rises steadily from 1.8% in 1950 to around 8% in 2023.

Figure 6. Externally Calibrated Time-Varying Parameters



Notes: The top-left and top-right panels plot final-producer markups, $1/\mu_j^y(t)$, and machine-producer markups, $1/\mu_j^x(t)$, respectively, for Goods, RI-Services, and NRI-Services. The middle-left and middle-right panels plot machine-producer productivity and labor-enhancing productivity, respectively, with both series normalized to one in 1950. In the first four panels, hollow diamonds, squares, and triangles denote observations for Goods, RI-Services, and NRI-Services, respectively, with marker area proportional to the corresponding sector's value-added share. The bottom-left panel plots the annual population growth rate, while the bottom-right panel plots the economy-wide share of R&D workers. Hollow hexagons denote economy-wide observations. All underlying data series are fitted using cubic smoothing splines with smoothing parameter 10^4 . For machine-producer and labor-enhancing productivity, the dashed lines are constant-growth paths constructed from the smoothed 1950 and 2023 endpoints. In the remaining panels, the dashed lines show the corresponding smoothed series. These dashed paths are the time-varying inputs used in the external calibration. RI-Services and NRI-Services denote research-intensive and non-research-intensive Services, respectively.

4.2 Internal Calibration

The remaining parameters are calibrated internally so that the model reproduces key features of U.S. structural change in production and innovation, as well as sectoral labor productivity growth, between 1950 and 2023. We impose a parsimonious time path for sectoral research productivity,

$$\zeta_j(t) = \zeta_j(1950) \exp [g_{\zeta,j}(t - 1950)], \quad j \in \{m, rs, ns\},$$

where $g_{\zeta,j}$ is sector specific and constant over time.

We also introduce two time-varying sectoral relative-price wedges, $\tau_{rs}^p(t)$ and $\tau_{ns}^p(t)$, with Goods as the reference sector. We recover these wedges period by period so that the model reproduces the observed paths of sectoral relative prices.¹⁹

Beyond these assumptions, we impose two normalizations. First, following [Comin et al. \(2021\)](#), we normalize one sectoral income-elasticity parameter to unity and set $\eta_{ns} = 1$. Second, because the overall scale of the sectoral consumption weights is not separately identified, we normalize them to sum to one, $\theta_m + \theta_{rs} + \theta_{ns} = 1$.

Thus, conditional on any two consumption weights, the third is determined residually.

These assumptions and normalizations leave 14 independent parameters to be calibrated internally:

$$\Theta = \{\sigma, \theta_m, \theta_{rs}, \eta_m, \eta_{rs}, \kappa_m, \kappa_{rs}, \kappa_{ns}, \zeta_m(1950), \zeta_{rs}(1950), \zeta_{ns}(1950), g_{\zeta,m}, g_{\zeta,rs}, g_{\zeta,ns}\}.$$

The parameters in Θ are jointly calibrated using a nested block-recursive simulated method of moments (SMM) routine. Conditional on a candidate parameter vector, we solve the model numerically and compare model-implied moments and paths with their empirical counterparts. The calibration combines exact recovery of the consumption weights and time-varying sectoral relative-price wedges, root-solving for the sectoral variety-expansion parameters κ_j , and SMM for the income-elasticity parameters, the elasticity of substitution across sectors, and the level and growth parameters governing sectoral research productivity. Within each block, the relevant parameters are chosen conditional on the parameters in the remaining blocks. We iterate across blocks until the parameter vector converges to a fixed point.²⁰

¹⁹We treat the relative-price wedges as residual calibration objects and do not assign them a structural interpretation. Matching sectoral relative prices is important for disciplining labor productivity because prices enter the capital-deepening component of sectoral productivity. Discrepancies between model-implied and observed relative prices would therefore translate directly into distorted capital-deepening dynamics, confounding the calibration of other channels affecting labor productivity.

²⁰Appendix I provides a complete description of the internal calibration procedure.

The results of the internal calibration are reported in Table 2. For each parameter, the table reports its calibrated value and the empirical moment or path that is most informative for its calibration. The sectoral variety-expansion parameters, κ_j , are primarily disciplined by the sectoral labor-productivity paths between 1950 and 2023. The larger value of κ_{ns} implies that, holding other determinants fixed, labor productivity in NRI-Services is more responsive to an expansion in the stock of machine varieties than in the other sectors.

The calibrated elasticity of substitution across sectors, σ , is close to zero, implying strong complementarity in sectoral demand.²¹ The parameter is disciplined by the joint evolution of sectoral relative prices and value-added shares over 1950–2023. The sectoral consumption weights are recovered analytically to match initial sectoral value-added shares, while the income-elasticity parameters are primarily disciplined by the subsequent evolution of these shares. Since $\sigma < 1$, higher values of η_j correspond to greater income elasticities. The calibrated values therefore imply that RI-Services is relatively income elastic and Goods relatively income inelastic, with NRI-Services providing the normalization $\eta_{ns} = 1$.

Table 2. Internally Calibrated Parameters

Parameter	Interpretation	Value	Most informative moment
κ_m	Goods variety-expansion parameter	[value]	Goods labor productivity path, 1950–2023
κ_{rs}	RI-Serv. variety-expansion parameter	0.934	RI-Serv. labor productivity path, 1950–2023
κ_{ns}	NRI-Serv. variety-expansion parameter	2.185	NRI-Serv. labor productivity path, 1950–2023
σ	Elasticity of substitution across sectors	0.031	Sectoral relative prices and value-added shares, 1950–2023
θ_m	Goods consumption weight	0.395	Goods value-added share, 1950
θ_{rs}	RI-Serv. consumption weight	0.104	RI-Serv. value-added share, 1950
η_m	Goods Engel parameter	0.818	Goods value-added share path, 1950–2023
η_{rs}	RI-Serv. Engel parameter	1.517	RI-Serv. value-added share path, 1950–2023
$\zeta_m(1950)$	Initial Goods research productivity	0.161	Goods labor productivity path, 1950–2023
$\zeta_{rs}(1950)$	Initial RI-Serv. research productivity	0.155	RI-Serv. scientist share path, 1950–2023
$\zeta_{ns}(1950)$	Initial NRI-Serv. research productivity	0.113	NRI-Serv. scientist share path, 1950–2023
$g_{\zeta,m}$	Goods research productivity growth	0.015	Goods labor productivity path, 1950–2023
$g_{\zeta,rs}$	RI-Serv. research productivity growth	0.021	RI-Serv. scientist share path, 1950–2023
$g_{\zeta,ns}$	NRI-Serv. research productivity growth	−0.019	NRI-Serv. scientist share path, 1950–2023

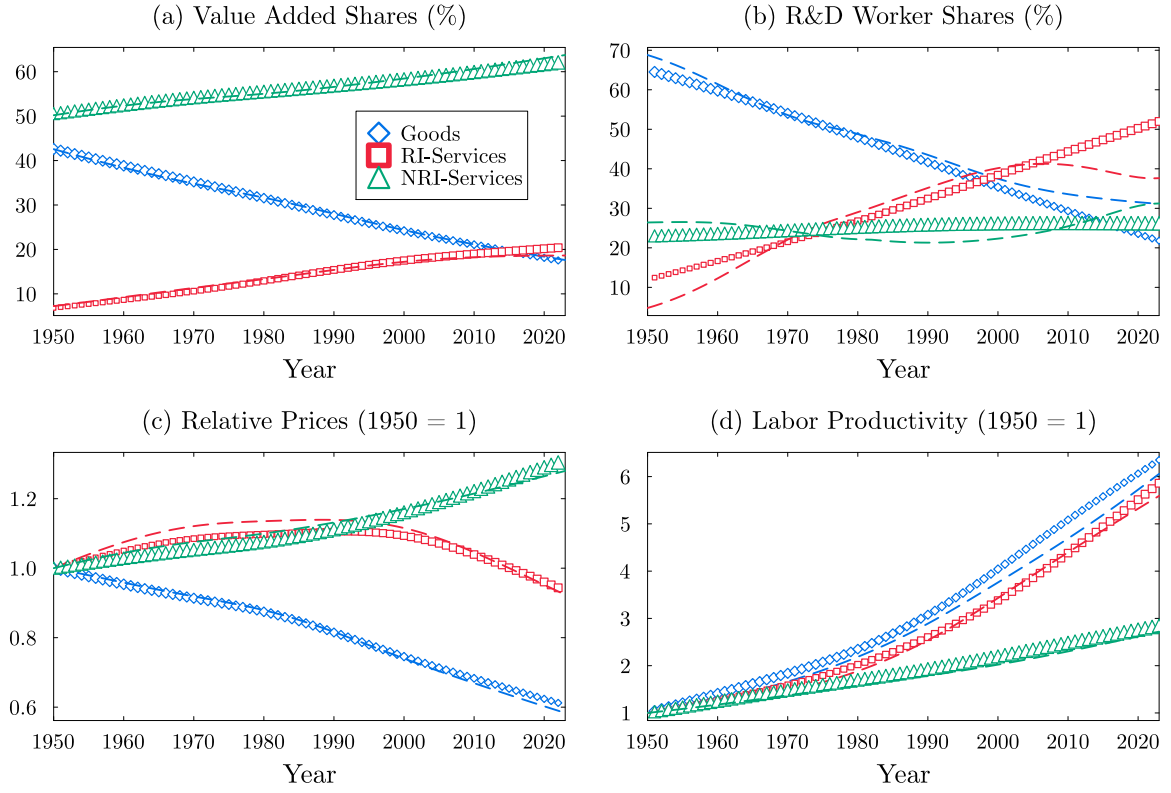
Notes: The table reports internally calibrated parameters. Parameters are calibrated jointly using the full set of calibration targets; the final column reports the moment or path that is most informative for each parameter. The consumption weights θ_m and θ_{rs} are disciplined by 1950 value-added shares, with $\theta_{ns} = 1 - \theta_m - \theta_{rs}$. The income-elasticity parameter for NRI-Services is normalized to $\eta_{ns} = 1$.

The technology parameters governing innovation and the productivity effects of new varieties are jointly calibrated using sectoral patent stocks, labor-productivity paths, and scientist shares. In the Goods sector, κ_m , $\zeta_m(1950)$, and $g_{\zeta,m}$ are jointly disciplined by the evolution of the Goods patent stock and the Goods labor-productivity path. These two targets separate the accumulation of ideas from their contribution to measured productivity and thereby pin down the level and growth of Goods research productivity.

²¹Low elasticities of substitution across broad sectors are common in quantitative models of structural change; see, for example, Buera & Kaboski (2009).

For the two service sectors, the research-productivity parameters are primarily disciplined by the allocation of scientists across sectors, while κ_{rs} and κ_{ns} are chosen to reproduce sectoral labor-productivity growth. These objects cannot be calibrated independently. Research productivity affects the allocation of scientists, the accumulation of ideas, and labor productivity in all sectors through knowledge spillovers and general-equilibrium effects. We therefore iterate over the research-productivity and variety-expansion parameters until the implied scientist shares and sectoral productivity paths jointly converge to their empirical counterparts. The demand parameters are subsequently re-estimated at the resulting technology paths, and the full procedure is iterated to a fixed point.²²

Figure 7. Internal Calibration Model Fit



Notes: The top-left panel plots sectoral value-added shares, while the top-right panel plots sectoral shares of R&D workers. Both are reported in percent. The bottom-left panel plots sectoral relative-price indices, while the bottom-right panel plots sectoral labor-productivity indices; both sets of indices are normalized to one in 1950. Hollow diamonds, squares, and triangles denote the smoothed data series for Goods, RI-Services, and NRI-Services, respectively, with marker area proportional to the corresponding sector's value-added share. Dashed lines are the model-simulated counterparts under the calibration reported in Table 2. RI-Services and NRI-Services denote research-intensive and non-research-intensive Services, respectively.

Figure 7 summarizes the model's fit to the targeted empirical paths. The top-left panel reports sectoral value-added shares and the top-right panel sectoral scientist shares. The

²²See Appendix I for further details on the internal calibration procedure.

bottom panels report sectoral relative prices and labor productivity, both normalized to one in 1950. The model closely reproduces the observed structural transformation in production and innovation, including the reallocation of researchers toward RI-Services and away from Goods, as well as the divergent labor-productivity paths of Goods and RI-Services relative to NRI-Services. The relative prices of RI-Services and NRI-Services are matched by construction through the time-varying relative-price wedges, whereas the Goods relative-price path is an untargeted implication of the model.

5 Quantitative Exercises

With the calibrated model in hand, we now use it to quantify the mechanisms underlying the joint evolution of structural change, innovation, and productivity. We begin by decomposing sectoral labor productivity growth into its underlying sources.

5.1 The Sources of Sectoral Productivity Growth

Proposition 1 expresses sectoral labor productivity as the product of endogenous innovation, machine intensity, and labor efficiency. Log-differentiating that expression yields the exact decomposition

$$g_{A_j}(t) = \underbrace{e_j^Y(t)g_{N_j}(t) + \dot{e}_j^Y(t) \ln N_j(t)}_{\text{Endogenous innovation}} + \underbrace{\frac{1-\alpha}{\alpha} \left[g_{z_j}(t) + g_{\mu_j^x}(t) \right]}_{\text{Embodied technical progress}} + \underbrace{\frac{1-\alpha}{\alpha} \left[g_{p_j}(t) + g_{\mu_j^y}(t) \right]}_{\text{Relative price effect}} + \underbrace{g_{\bar{h}_j}(t)}_{\text{Labor efficiency}}, \quad (18)$$

where $e_j^Y(t)$ is defined in Proposition 1. We use this decomposition to quantify the sources of the sectoral productivity growth differences documented in Section 2.²³ Table 3 reports the annualized contribution of each channel over 1950–2023. The corresponding decompositions for 1950–1980 and 1980–2023 are reported in Appendix H.

Over 1950–2023, labor productivity grows by 2.47 percent per year in Goods, 2.36 percent in RI-Services, and 1.37 percent in NRI-Services Table 3 shows that the large productivity-growth differential between Goods and NRI-Services originates primarily in endogenous innovation.

Endogenous innovation contributes 2.47 percentage points to annual productivity growth in Goods, compared with 1.10 percentage points in RI-Services and 0.56 percentage points in NRI-Services The resulting 1.91 percentage point difference between Goods and NRI-

²³In discrete time, we use the midpoint decomposition for the product $e_{Y,j}(t) \ln N_j(t)$, so that changes in the variety stock and changes in its productivity elasticity add exactly to the innovation contribution.

Table 3. Decomposition of Sectoral Labor Productivity Growth, 1950–2023

	Sector		
	Goods	RI-Serv.	NRI-Serv.
I. Endogenous innovation	2.47	1.10	0.56
(a) <i>Idea growth</i>	2.65	1.37	0.59
(b) <i>Elasticity drift</i>	−0.18	−0.28	−0.03
II. Embodied technical progress	0.07	0.78	0.19
(a) <i>Machine productivity</i>	−0.06	0.62	0.10
(b) <i>Machine market power</i>	0.13	0.16	0.09
III. Relative price effect	−0.44	0.07	0.24
(a) <i>Output price</i>	−0.32	−0.05	0.14
(b) <i>Sectoral market power</i>	−0.13	0.12	0.09
IV. Labor efficiency	0.38	0.41	0.39
Total	2.47	2.36	1.37

Notes: Entries report annualized contributions, in percentage points, to the growth of sectoral labor productivity $\mathcal{A}_j = Y_j/L_j^p$ over 1950–2023. Endogenous innovation is decomposed into growth in the stock of machine varieties and changes in its productivity elasticity. Embodied technical progress combines machine productivity and machine market power. The relative price effect combines changes in sectoral output prices and sectoral market power. Components may not sum exactly because of rounding.

Services exceeds their observed productivity growth differential of 1.10 percentage points. Other forces therefore compress the productivity gap generated by innovation. In particular, the relative price channel reduces Goods productivity growth by 0.44 percentage points while raising NRI-Services productivity growth by 0.24 percentage points. Labor efficiency contributes almost equally across sectors, between 0.38 and 0.41 percentage points, and consequently contributes little to their productivity-growth differences.

RI-Services provides a useful comparison. Its productivity growth is almost as high as that of Goods despite receiving less than half the contribution from endogenous innovation, 1.10 versus 2.47 percentage points. The difference is largely offset by embodied technical progress, which contributes 0.78 percentage points in RI-Services and only 0.07 percentage points in Goods. Similar productivity growth in these two sectors therefore reflects different underlying sources of technical change.

These results identify endogenous innovation as the main source of the productivity-growth differential associated with the Baumol mechanism in our model. The slow productivity growth of NRI-Services reflects its weak accumulation of new varieties, while the considerably stronger innovation response in Goods generates a productivity advantage that relative-price movements partially offset.²⁴

²⁴The subperiod decomposition in Appendix H shows that this conclusion is not driven by averaging across the postwar period. In Goods, the contribution of endogenous innovation rises from 2.30 per-

The importance of endogenous innovation in Table 3 raises the next question: why does the accumulation of varieties differ so sharply across sectors? We turn to this question by decomposing the flow of new ideas into research productivity, research employment, accumulated knowledge, and the changing structure of cross-sector knowledge spillovers.

5.2 The Sources of Idea Growth

Exercise 1 shows that differences in idea accumulation are the main source of the productivity growth differential between Goods and NRI-Services. We now examine why innovation evolves so differently across sectors. We study the flow of new varieties, $\dot{N}_j(t)$, which is the direct output of the innovation technology.

Taking logs of Equation (5) gives

$$\ln \dot{N}_j(t) = \ln \zeta_j(t) + \phi \ln L_j^s(t) + \gamma \sum_i \delta_{ji}(t) \ln \tilde{N}_i(t), \quad (19)$$

where $\tilde{N}_i(t) = \bar{N}_i N_i(t)$ incorporates inherited knowledge and subsequent endogenous idea accumulation, as described in Section 4.²⁵

Differencing Equation (19) across adjacent years and applying the exact midpoint decomposition to the knowledge-spillover term yields

$$\begin{aligned} \Delta_t \ln \dot{N}_j = & \underbrace{\Delta_t \ln \zeta_j}_{\text{Research productivity}} + \underbrace{\phi \Delta_t \ln L_j^s}_{\text{Research employment}} + \underbrace{\gamma \bar{\delta}_{jj,t} \Delta_t \ln N_j}_{\text{Spillovers from own accumulated ideas}} \\ & + \underbrace{\gamma \sum_{i \neq j} \bar{\delta}_{ji,t} \Delta_t \ln N_i}_{\text{Spillovers from other sectors' accumulated ideas}} + \underbrace{\gamma \sum_i \ln \bar{N}_i \Delta_t \delta_{ji}}_{\text{Spillover reallocation, inherited knowledge}} \\ & + \underbrace{\gamma \sum_i \ln \bar{N}_{it} \Delta_t \delta_{ji}}_{\text{Spillover reallocation, accumulated ideas}}, \end{aligned} \quad (20)$$

where

$$\Delta_t x \equiv x(t) - x(t-1), \quad \bar{x}_t \equiv \frac{x(t) + x(t-1)}{2}.$$

The decomposition follows from $\tilde{N}_i(t) = \bar{N}_i N_i(t)$, so that changes in effective knowl-

centage points in 1950–1980 to 2.59 percentage points in 1980–2023, even as overall productivity growth falls from 2.61 to 2.37 percent. The decline instead reflects a more negative relative price effect, whose contribution falls from -0.13 to -0.66 percentage points. The post-1980 slowdown in Goods therefore occurs despite a larger contribution from innovation.

²⁵Recall that the constants $\bar{N}_i$, disciplined by sectoral patent shares in 1950, determine the relative size of sectoral knowledge stocks at the beginning of the sample, while the production-side variety stocks are normalized to $N_i(1950) = 1$.

edge reflect endogenous idea accumulation, while $\bar{N}_i$ captures inherited differences in knowledge stocks. The six terms separate changes in the flow of new ideas into research productivity, research employment, accumulated knowledge, and changes in the direction of knowledge spillovers. Summing Equation (20) over 1951–2023 gives an exact decomposition of $\ln \dot{N}_j(2023) - \ln \dot{N}_j(1950)$. The results of this decomposition, by broad sector, are presented in Table 4 below.

Table 4. Decomposition of the Growth in the Flow of New Ideas, 1950–2023

	Sector		
	Goods	RI-Serv.	NRI-Serv.
I. Research productivity	1.07	1.55	−1.39
II. Research employment	0.80	2.22	1.28
III. Own accumulated ideas	0.97	0.18	0.04
IV. Other sectors’ accumulated ideas	0.03	0.81	0.84
V. Spillover reallocation, inherited knowledge	−0.26	−0.51	−0.29
VI. Spillover reallocation, accumulated ideas	−0.03	−0.06	0.00
Total	2.58	4.19	0.47
<i>Memo:</i>			
Flow of new ideas, 1950	0.017	0.003	0.005
Flow of new ideas, 2023	0.230	0.221	0.009
Cumulative stock growth, $\ln N_j(2023) - \ln N_j(1950)$	2.02	1.87	0.42

Notes: Rows I–VI report contributions, in log points, to the change in the log flow of new varieties, $\ln \dot{N}_j(2023) - \ln \dot{N}_j(1950)$. The decomposition follows Equation (20) and is exact before rounding. The memo rows report the initial and final flow of new varieties and cumulative growth in the stock of varieties.

RI-Services exhibits the largest expansion in innovation, with the flow of new ideas rising by 4.19 log points. The dominant contribution comes from research employment, which adds 2.22 log points and reflects the large reallocation of scientists toward the sector documented in Section 2. Research productivity contributes a further 1.55 log points. Knowledge accumulation also plays an important role, but primarily through spillovers from outside the sector: ideas accumulated in other sectors contribute 0.81 log points, compared with only 0.18 from RI-Services’s own accumulated knowledge. The expansion of innovation in RI-Services therefore combines a strong increase in research inputs with substantial use of knowledge generated elsewhere in the economy.

NRI-Services displays a very different pattern. Research employment contributes 1.28 log points and spillovers from knowledge accumulated in other sectors contribute another 0.84 log points. Taken together, these forces would imply substantial growth in the flow

of new ideas. They are instead largely offset by declining research productivity, which contributes -1.39 log points. As a result, the flow of new ideas rises by only 0.47 log points over the sample, and the stock of varieties expands by just 0.42 log points. Thus, the weak accumulation of ideas in NRI-Services does not reflect an absence of research labor or access to external knowledge. In the calibrated model, it is the decline in $\zeta_{ns}(t)$ that reconciles these favorable forces with the sector’s limited innovation growth.

The accumulated-knowledge terms reveal a broader asymmetry in how sectors build on existing ideas. Goods draws almost entirely on its own expanding knowledge stock: spillovers from its own accumulated ideas contribute 0.97 log points, compared with only 0.03 from other sectors. The two service sectors exhibit the reverse pattern. In both RI-Services and NRI-Services, knowledge accumulated elsewhere contributes substantially more to innovation growth than knowledge accumulated within the sector itself. Innovation in Services is therefore considerably more dependent on the evolution of the economy-wide knowledge base, whereas innovation in Goods is more closely tied to its own past accumulation of ideas.²⁶

Changes in spillover weights make negative contributions in all three sectors. Most of this effect operates through inherited knowledge rather than through differences in idea accumulation after 1950. Since inherited knowledge is initially concentrated in Goods, changes in the $\delta_{ji}(t)$ weights that reduce exposure to sectors with larger inherited knowledge stocks lower the effective knowledge input entering research. By contrast, the interaction between changing spillover weights and post-1950 idea accumulation is quantitatively small. We therefore keep the inherited-knowledge and accumulated-knowledge components separate in Table 4.

Taken together, the two decompositions clarify the source of sectoral productivity divergence in the model. Differences in labor-productivity growth are driven primarily by unequal accumulation of machine varieties, but the forces behind that accumulation differ sharply across sectors. Goods innovation is supported by research productivity, research employment, and the expansion of its own knowledge stock. RI-Services benefits from a large reallocation of scientists and from knowledge generated elsewhere. NRI-Services, despite receiving additional researchers and substantial external knowledge spillovers, accumulates relatively few new varieties because these forces are offset by declining research productivity. The counterfactuals that follow quantify how much of these endogenous innovation paths can be traced to structural change.

²⁶This asymmetry will be important in the counterfactuals because reallocating research effort changes both current idea production and the knowledge stocks available for subsequent innovation.

5.3 Structural Change and Aggregate Productivity

We now examine how structural change affects aggregate labor productivity. Structural change can operate through two conceptually distinct margins. First, reallocating production across sectors changes aggregate productivity even if sectoral productivity paths are held fixed. This is the conventional accounting channel emphasized in the literature linking structural change to aggregate productivity slowdowns. Second, in our model, changes in sectoral demand also affect machine use and innovation incentives, so sectoral productivity responds endogenously.

We measure aggregate labor productivity directly from the model as real value added per production worker. Since aggregate value added equals final output net of the final good used in machine production, the resource constraint implies

$$\mathcal{A}^{VA}(t) \equiv \frac{VA(t)}{L^p(t)} = \frac{C(t)}{L^p(t)}, \quad L^p(t) = \sum_j L_j^p(t). \quad (21)$$

We use Equation (21) to evaluate aggregate productivity in the model and in each counterfactual.

To isolate the conventional composition effect, we also construct a Törnqvist accounting index using the sectoral value-added shares defined in Section 3

$$\Delta \ln \left(\frac{VA(t)}{L^p(t)} \right) = \sum_j \bar{s}_j^{VA}(t) \Delta \ln [\omega_j(t) \mathcal{A}_j(t) L_j^p(t)] - \Delta \ln L^p(t), \quad (22)$$

where $\bar{s}_j^{VA}(t)$ is the average value-added share between adjacent years and $\omega_j(t)$ converts sectoral gross output into value added. This accounting index holds sectoral productivity paths fixed when measuring the composition effect, whereas the model counterfactuals allow $\mathcal{A}_j(t)$ to respond endogenously.

To isolate the forces driving changes in sectoral value-added shares, we construct counterfactuals that sequentially remove three channels. We eliminate income effects by setting $\eta_j = 1$ for all sectors, remove the response of expenditure shares to relative prices by setting $\sigma = 1$, and eliminate markup drift by holding $\mu_j^y(t)$ and $\mu_j^x(t)$ at their 1950 values. Together, these restrictions hold expenditure shares and the value-added wedges fixed over time.²⁷

The contribution attributed to each force depends on the order of the shutdowns. Because setting $\sigma = 1$ also eliminates income effects from expenditure shares, we remove income effects before relative-price effects and treat these two shutdowns as an ordered

²⁷In addition, in each counterfactual we re-solve the expenditure weights θ_j to reproduce baseline 1950 value-added shares, while prices, production employment, and the allocation of researchers adjust in equilibrium.

pair. We then remove markup drift either before or after the pair. For each outcome, we compute each force’s marginal contribution under these two sequences and report their average. The contributions sum to the difference between the baseline and full-shutdown outcomes. Appendix ?? reports the associated cumulative accounting results. Its intermediate columns subtract these average contributions from the baseline and do not represent separately solved equilibria.

For comparison, the fixed-shares benchmark is

$$g^{\text{fixed}}(t) = \sum_j s_j^{VA}(1950) \Delta \ln \mathcal{A}_j^{\text{baseline}}(t).$$

It applies initial value added shares to baseline sectoral productivity growth. Comparing this benchmark with the equilibrium counterfactual shows how the assessment changes when sectoral productivity responds endogenously to the shutdowns.

Table 5. Structural Change and Aggregate Labor Productivity

	Avg. annual growth (%) 1950–2023	Productivity index, 2023 1950 = 100
Baseline	1.77	362.7
Fixed-shares benchmark	1.90	399.9
Full shutdown	2.13	474.7

Notes: Aggregate labor productivity is real value added per production worker. Average annual growth is the annualized log change over 1950–2023. Productivity indices are normalized to 100 in 1950. The fixed-shares benchmark weights baseline sectoral labor productivity growth by 1950 value added shares; its endpoint index is approximated from the reported cumulative log growth. The full shutdown sets $\eta_j = 1$ and $\sigma = 1$ and holds both markup paths at their 1950 values, with the equilibrium re-solved.

Table 5 shows that holding sectoral value-added shares at their 1950 levels raises aggregate labor productivity by about 10% by 2023 relative to the baseline. In annual growth terms, aggregate productivity grows at 1.90 percent under the fixed-shares benchmark, compared with 1.77 percent in the baseline, a difference of 0.13 percentage points per year. The conventional accounting approach therefore attributes a quantitatively meaningful share of the aggregate productivity slowdown to the reallocation of activity across sectors. The full equilibrium shutdown implies a substantially larger effect: aggregate labor productivity is about 31% higher by 2023 than in the baseline. In annual growth terms, productivity grows at 2.13 percent under the full shutdown, 0.37 percentage points faster than in the baseline.

Table 6 traces the mechanisms underlying the larger equilibrium effect. The response of expenditure shares to relative prices is the dominant channel, accounting for about 64 percent of the cumulative productivity gap between the baseline and the full shutdown. Its marginal contribution is equivalent to 19.7 percent of baseline aggregate labor produc-

Table 6. Mechanisms Behind Structural Change and Labor Productivity

	Baseline	Baseline minus income contribution	Baseline minus income and relative-price contributions	Full shutdown
<i>Panel A. Aggregate labor productivity, 2023 (1950 = 100)</i>				
Aggregate	362.7	379.6	450.9	474.7
<i>Panel B. Sectoral value-added shares, 2023 (%)</i>				
Goods	18.1	23.6	45.6	41.4
RI-Serv.	18.0	7.8	7.0	7.4
NRI-Serv.	63.9	68.6	47.5	51.2
<i>Panel C. Sectoral scientist shares, 2023 (%)</i>				
Goods	31.1	47.9	81.6	81.9
RI-Serv.	37.6	12.0	5.4	10.6
NRI-Serv.	31.2	40.1	13.0	7.5
<i>Panel D. Machine-variety indices, 2023 (1950 = 1)</i>				
Goods	7.509	9.189	11.534	11.730
RI-Serv.	6.477	4.705	3.919	3.942
NRI-Serv.	1.521	1.565	1.401	1.365
<i>Panel E. Labor-productivity indices, 2023 (1950 = 1)</i>				
Goods	6.058	7.242	9.110	10.037
RI-Serv.	5.593	5.198	5.222	5.601
NRI-Serv.	2.720	2.880	2.852	2.698

Notes: The table reports selected outcomes from the cumulative decomposition of the structural-change experiment. The first intermediate column subtracts the average marginal contribution of income effects from the baseline; the second additionally subtracts the average marginal contribution of the response of expenditure shares to relative prices. These intermediate columns are accounting constructs and do not represent separately solved equilibria. The full shutdown sets $\eta_j = 1$ and $\sigma = 1$ and holds $\mu_j^y(t)$ and $\mu_j^x(t)$ at their 1950 values. Appendix ?? reports the full sectoral decomposition.

tivity in 2023 (379.6 to 450.9 in Panel A). By comparison, income effects account for 4.7 percent of baseline productivity and markup drift for 6.6 percent, together yielding the approximately 31 percent gap between the baseline and the full-shutdown equilibrium.

The mechanism operates through sectoral expenditure shares. Because $\sigma < 1$, expenditure shares move positively with relative prices: faster productivity growth in Goods lowers its relative price and, in turn, reduces its share of expenditure. To isolate this channel, we first remove income effects and then set $\sigma = 1$, which makes expenditure shares invariant to relative prices. The experiment does not hold relative prices fixed. Prices continue to adjust endogenously and continue to affect machine demand, capital deepening, and the value of innovation; what is removed is only their effect on the allocation of final expenditure across sectors.

This channel is particularly important for Goods. Subtracting the relative-price contribution raises its 2023 value-added share from 23.6 to 45.6 percent, while the NRI-Services share falls from 68.6 to 47.5 percent. The associated change in market size also redirects research toward Goods, so the productivity effect extends beyond the contemporaneous reallocation of production.

Income effects operate differently, primarily by supporting the expansion of RI-Services. Subtracting their contribution lowers its 2023 value-added share from 18.0 to 7.8 percent and its scientist share from 37.6 to 12.0 percent, while raising the Goods scientist share from 31.1 to 47.9 percent. The income channel therefore affects aggregate productivity not only through sectoral reallocation, but also by changing the market sizes that direct research across sectors. Quantitatively, it accounts for about 15 percent of the cumulative productivity gap between the baseline and the full shutdown, equivalent to 4.7 percent of baseline aggregate productivity in 2023 (362.7 to 379.6 in Panel A).

These reallocations become much larger when all three channels are removed jointly. Under the full shutdown, the Goods value-added share is 41.4 percent in 2023, compared with 18.1 percent in the baseline, while its scientist share rises from 31.1 to 81.9 percent. Because the aggregate share of employment devoted to research is unchanged, this is fully a reallocation of research toward Goods. The larger Goods market raises demand for sector-specific machines and increases the value of developing new varieties.

The resulting increase in Goods innovation is reinforced by knowledge accumulation. As shown in Section 5.2, Goods innovation depends predominantly on its own accumulated knowledge stock. Redirecting research toward Goods therefore raises current idea production and expands the knowledge base available for subsequent innovation. By 2023, the Goods stock of machine varieties is 56 percent larger under the full shutdown than in the baseline, while Goods labor productivity is approximately 66 percent higher. The corresponding changes in service-sector labor productivity are small, despite sizable changes

in their variety stocks. The endogenous productivity response is therefore concentrated in Goods.

Markup drift accounts for the remaining 21 percent of the cumulative productivity gap, equivalent to 6.6 percent of baseline aggregate productivity in 2023 (450.9 to 474.7 in Panel A). Subtracting its contribution also returns sectoral value-added shares to their 1950 levels. At this stage, however, the associated reallocation of research is small: the Goods scientist share rises only from 81.6 to 81.9 percent. The productivity contribution of markup drift therefore operates primarily through margins other than the sectoral allocation of researchers.

The exercise therefore extends the conventional Baumol mechanism linking structural change to the slowdown in aggregate productivity growth. Structural change lowers aggregate productivity not only by reallocating activity toward sectors with slower productivity growth, but also by changing the productivity paths themselves. In the calibrated model, the contraction of the Goods market redirects research away from a sector in which innovation builds strongly on its own accumulated knowledge, weakening subsequent idea accumulation and productivity growth. Appendix ?? reports the full sectoral decomposition.

5.4 The Direction of Research and the Innovation Technology

The previous exercise shows that structural change affects the allocation of research and the resulting paths of sectoral productivity. We now examine which features of the innovation technology shape that allocation. The motivating observation is that production employment shifts from Goods toward both service sectors, while the reallocation of research employment is concentrated disproportionately in RI-Services. As a result, NRI-Services absorbs a large share of the expansion in production while attracting comparatively little research. We ask whether differences in research productivity and access to accumulated knowledge can account for this divergence in the allocation of production and innovation.

We consider three dimensions of heterogeneity in the innovation technology. First, sectors differ in their initial research productivity, $\zeta_j(1950)$. Second, they differ in its subsequent growth: calibrated research productivity grows by 1.47 percent per year in Goods and 2.13 percent in RI-Services, while declining by 1.90 percent in NRI-Services. Third, the spillover matrix $\delta_{ji}(t)$ gives sectors different access to their own and other sectors' accumulated knowledge. We homogenize each dimension in turn and then consider a counterfactual in which all three are homogenized jointly.

To homogenize knowledge spillovers, we give all sectors the same access to own and external knowledge. We set each sector's weight on its own accumulated knowledge equal

to the cross-sector average,

$$\bar{\delta}(t) = \frac{1}{3} \sum_j \delta_{jj}(t),$$

and divide the remaining weight, $1 - \bar{\delta}(t)$, equally across the other two sectors. This preserves the average importance of own knowledge and its evolution over time, while eliminating cross-sector differences in access to accumulated knowledge.²⁸

To homogenize research productivity, we impose either a common initial level or a common growth rate across sectors, choosing the common parameter to match the baseline 2023 value of the value-added-weighted Törnqvist index of normalized idea stocks. This holds the aggregate idea index fixed while allowing its sectoral composition, and hence aggregate labor productivity, to change. Because the symmetric-spillover experiment has no free research-productivity parameter, the aggregate idea index is allowed to respond. When the initial level and growth rate of research productivity are homogenized jointly, we average outcomes across the two possible orders of imposing these restrictions.²⁹

Table 7 shows that differences in the growth of research productivity are an important force behind the disproportionate allocation of research toward RI-Services rather than NRI-Services. Imposing a common growth rate raises the 2023 NRI-Services share of researchers from 31.2 to 45.0 percent and lowers the RI-Services share from 37.6 to 28.0 percent. Annual idea-stock growth correspondingly rises from 0.58 to 1.27 percent in NRI-Services and falls from 2.56 to 1.29 percent in RI-Services. As a result, the gap in labor productivity growth between the two service sectors narrows from 0.99 to 0.13 percentage points per year.

Differences in initial research productivity have smaller effects. Imposing a common initial level leaves the allocation of researchers close to the baseline: the 2023 shares change by approximately 0.1 percentage points in Goods, 1.4 points in RI-Services, and -1.4 points in NRI-Services. The contrast with the common-growth experiment indicates that the evolution of research productivity over time matters more for the 2023 allocation of research than differences in its initial level.

The spillover experiment reveals a different mechanism. Homogenizing access to accu-

²⁸To illustrate the magnitude of this adjustment, consider the 2023 elasticities. Since the elasticity of new-idea production in sector j with respect to knowledge accumulated in sector i is $\gamma\delta_{ji}$, the baseline own-knowledge elasticity is approximately 0.44 in Goods, 0.14 in RI-Services, and 0.08 in NRI-Services. The symmetric specification sets all three to approximately 0.22. Correspondingly, the elasticity with respect to the combined knowledge stock of the other two sectors rises from about 0.06 to 0.28 in Goods, while falling from 0.36 to 0.28 in RI-Services and from 0.42 to 0.28 in NRI-Services.

²⁹The aggregate idea-stock target provides only one normalization for the two common research-productivity parameters. We therefore consider both orders. In each case, the parameter homogenized first is fixed at its value from the corresponding single-channel experiment, while the second is chosen to match the baseline aggregate idea index. All other model parameters and calibrated paths retain their baseline values.

Table 7. The Direction of Research and the Innovation Technology

	Baseline	Symmetric spillovers	Common ζ_0	Common g_ζ	All three
<i>Panel A. Share of researchers in 2023 (%)</i>					
Goods	31.1	54.6	31.2	27.0	27.2
RI-Services	37.6	38.5	39.0	28.0	33.0
NRI-Services	31.2	6.9	29.8	45.0	39.8
<i>Panel B. Average annual idea-stock growth (%)</i>					
Goods	2.76	1.41	2.52	1.93	1.70
RI-Services	2.56	1.03	2.35	1.29	1.15
NRI-Services	0.58	0.11	0.68	1.27	1.45
<i>Panel C. Average annual labor productivity growth (%)</i>					
Goods	2.47	1.44	2.32	1.97	1.83
RI-Services	2.36	1.72	2.30	2.00	1.96
NRI-Services	1.37	0.88	1.44	1.87	1.99
<i>Memo: Aggregate idea index, 2023 (1950 = 100)</i>					
	300.1	163.4	300.1	300.1	300.1

Notes: Growth rates are annualized log changes over 1950–2023. Each single-channel experiment homogenizes the indicated feature of the innovation technology while retaining the other two at baseline. In the common- ζ_0 and common- g_ζ experiments, the common parameter is chosen to match the baseline 2023 aggregate idea index. The symmetric-spillover experiment does not impose this normalization. The final column imposes all three restrictions and reports average outcomes across the two orders of homogenizing the initial level and growth rate of research productivity. Research shares may not sum to 100 because of rounding.

mulated knowledge raises the Goods share of researchers to 54.6 percent and reduces the NRI-Services share to 6.9 percent. Despite this substantial reallocation toward Goods, idea growth falls in every sector, and the aggregate idea index reaches only 163.4, compared with 300.1 in the baseline. The reason is that Goods plays a central role in the baseline knowledge network: its innovation builds strongly on its own accumulated ideas, while both service sectors draw importantly on knowledge generated in Goods. Homogenizing spillovers weakens both margins. Goods becomes less able to build on its own knowledge stock, while the service sectors lose part of their access to Goods knowledge. The resulting loss in knowledge propagation more than offsets the larger share of researchers allocated to Goods.

The spillover experiment reinforces the earlier result that research employment alone is insufficient to assess innovation performance. The return to research depends on the knowledge stock on which it builds and on how newly generated knowledge propagates across sectors.

When all three dimensions of the innovation technology are homogenized jointly, the 2023 share of researchers in NRI-Services rises from 31.2 to 39.8 percent. Annual idea-stock

growth rises to 1.45 percent in NRI-Services and falls to 1.70 percent in Goods and 1.15 percent in RI-Services. Sectoral labor productivity growth consequently becomes much more similar, at 1.83 percent in Goods, 1.96 percent in RI-Services, and 1.99 percent in NRI-Services. Because the aggregate idea index is held at its baseline endpoint, this convergence reflects a redistribution of innovation across sectors rather than a change in aggregate idea accumulation.

The individual experiments also operate in different directions. Homogenizing research-productivity growth shifts research toward NRI-Services, whereas homogenizing knowledge spillovers shifts it strongly toward Goods. These offsetting forces help explain why jointly homogenizing the innovation technology produces smaller changes in the research allocation than the spillover experiment alone.

5.5 Composition, Aggregation, and the Innovation Feedback

Section 5.3 shows that the full shutdown raises cumulative aggregate labor productivity growth by 26.9 log points between 1950 and 2023, leaving productivity about 31 percent higher than in the baseline by 2023. We now ask how much of this gap reflects conventional composition effects and how much arises because sectoral productivity responds endogenously. We then decompose the productivity response into the innovation and capital-deepening channels identified above. The exercise compares the baseline and full-shutdown equilibrium paths.

To organize the cumulative productivity gap, we use the Törnqvist accounting decomposition introduced in Section 5.3. For each adjacent pair of years,

$$\Delta \ln \left(\frac{VA(t)}{L^p(t)} \right) = \underbrace{\sum_j \bar{s}_j^{VA}(t) \Delta \ln \mathcal{A}_j(t)}_{\text{Within-sector productivity}} + \underbrace{\sum_j \bar{s}_j^{VA}(t) \Delta \ln \left(\frac{L_j^p(t)}{L^p(t)} \right)}_{\text{Labor reallocation}} + \underbrace{\sum_j \bar{s}_j^{VA}(t) \Delta \ln \omega_j(t)}_{\text{Value-added conversion}}. \quad (23)$$

We sum each term over 1950–2023 to obtain its contribution to cumulative aggregate productivity growth. The first term aggregates sectoral productivity growth, while the second captures changes in the allocation of production employment. The third captures changes in the conversion of sectoral gross revenue into value added induced by the markup paths. It is zero under the full shutdown because $\mu_j^y(t)$ and $\mu_j^x(t)$, and hence $\omega_j(t)$, are held fixed at their 1950 values.

The difference in the first term across equilibria reflects both changes in sectoral productivity growth and changes in the weights applied to that growth. To separate them, let $g_j^r(t) = \Delta \ln \mathcal{A}_j^r(t)$ denote sectoral productivity growth in equilibrium $r \in \{B, F\}$, where B denotes baseline and F denotes full shutdown. Let $w_j^r(t) = \bar{s}_j^{VA,r}(t)$ denote the corresponding Törnqvist weight. At each date, the exact midpoint identity gives

$$\sum_j w_j^F g_j^F - \sum_j w_j^B g_j^B = \underbrace{\sum_j \frac{w_j^F + w_j^B}{2} (g_j^F - g_j^B)}_{\text{Productivity feedback}} + \underbrace{\sum_j \frac{g_j^F + g_j^B}{2} (w_j^F - w_j^B)}_{\text{Reweighting}}. \quad (24)$$

where time arguments are suppressed.

The productivity-feedback term measures changes in sectoral productivity growth using weights averaged across the two equilibria. We group the reweighting term with changes in the labor-reallocation and value-added-wedge terms to obtain the accounting component. This decomposition allocates the interaction between changing weights and changing productivity symmetrically across the two components.

The accounting component differs from the fixed-initial-shares benchmark in Section 5.3. That benchmark applies 1950 weights to baseline sectoral productivity growth. Here, the accounting component uses both equilibrium paths and includes changes in labor reallocation and value added wedges. The two calculations address related questions but need not produce identical numbers.

Table 8. Composition and Cumulative Aggregate Labor Productivity Growth

	Baseline	Full shutdown	Difference
<i>Panel A. Cumulative aggregate productivity growth</i>			
I. Within-sector productivity	134.5	159.1	24.6
II. Labor reallocation	0.7	−3.4	−4.0
III. Value-added conversion	−6.4	0.0	6.4
Total	128.8	155.8	26.9
<i>Panel B. Decomposition of the total difference</i>			
I. Accounting component	—	—	10.2
II. Productivity feedback	—	—	16.6
Total	—	—	26.9
<i>Panel C. Decomposition of productivity feedback</i>			
(a) Endogenous innovation	—	—	18.6
(b) Embodied technical progress	—	—	−8.5
(c) Relative-price effect	—	—	6.5
(d) Labor efficiency	—	—	0.0
Total	—	—	16.6

Notes: Entries are cumulative log-point contributions over 1950–2023. Difference is full shutdown minus baseline. Panel A follows Equation (23). Panel B separates changes in sectoral productivity growth from reweighting using Equation (24); the accounting component combines reweighting with changes in the labor-reallocation and value-added-conversion terms. Panel C applies the sectoral productivity decomposition in Equation (18) to the productivity-feedback component. Differences and totals are calculated before rounding.

Panel A of Table 8 shows that the within-sector contribution rises from 1.84 to 2.18 percentage points per year under the full shutdown. Changes in labor reallocation subtract 0.06 percentage points from the gain, while eliminating changes in the value added wedge adds 0.09 percentage points. Constant value added shares therefore do not eliminate the labor-reallocation term: production employment shares continue to adjust in the counterfactual equilibrium.

Panel B separates the increase in within-sector productivity contributions into changes in weights and changes in sectoral productivity growth. The accounting component contributes 10.2 log points to the cumulative productivity gain, while the productivity feedback contributes 16.6 log points. Approximately 62 percent of the aggregate productivity difference therefore comes from the response of sectoral productivity. Most of the effect of structural change on aggregate productivity thus operates through changes in sectoral productivity paths rather than through composition alone.

Panel C decomposes the 16.6-log-point productivity feedback into its underlying sectoral productivity channels. Endogenous innovation is the largest positive component, contributing 18.6 log points through changes in the accumulation of machine varieties and their contribution to sectoral productivity. This response is driven in part by the stronger accumulation of ideas in Goods under the full shutdown documented in Section 5.3. The relative-price effect contributes a further 6.5 log points, while embodied technical progress offsets 8.5 log points of these gains. Labor efficiency makes no contribution because its path is unchanged across the two equilibria.

The labels in Panel C follow the sectoral productivity decomposition developed above. Embodied technical progress combines machine productivity and machine-producer market power, while the relative-price effect combines sectoral output prices and output-producer market power. Because the exogenous machine-productivity paths are unchanged across equilibria, differences in the embodied-technical-progress contribution arise purely from the endogenous markup component. Taken together, the two capital-deepening channels make a small negative net contribution to the productivity feedback.

The value-added-conversion term in Panel A is distinct from the contribution assigned to the markup shutdown above. The former isolates the direct accounting effect of changes in $\omega_j(t)$, whereas the latter includes all equilibrium responses to fixing markups, including changes in prices, allocations, and innovation. The decomposition therefore separates the channels through which aggregate productivity changes from the counterfactual interventions that generate those changes.

Taken together, the results show that structural change affects aggregate productivity through both composition and endogenous sectoral productivity. Of the 26.9-log-point cumulative productivity gap between the baseline and the full shutdown, 16.6 log points,

or about 62 percent, reflect changes in sectoral productivity paths, compared with 10.2 log points from the accounting component. Within the productivity response, endogenous innovation contributes 18.6 log points, partly offset by a net contribution of -2.0 log points from the capital-deepening channels.

6 Conclusion

How has structural change shaped postwar US aggregate productivity growth? A standard fixed-shares exercise implies that aggregate productivity would have grown by 1.90% compared to 1.77% per year in the baseline. In this paper, we argue that this calculation overlooks the circular link between structural change and productivity growth; structural change not only shapes the sectoral composition of production activity, it also determines the allocation of innovation activity in the economy. Consequently, any evaluation of the role that structural change has played in shaping productivity growth requires accounting for how structural change has shaped this growth at the sector level.

We motivate this claim using postwar US data on sectoral innovation patterns. We show that while innovation activity – measured by R&D workers and patents – has relocated away from Goods and into Services, much of this relocation has been concentrated in RI-Services. Notably, RI-Services productivity growth has largely kept pace with the rate in Goods sector, while that of NRI-Services has lagged behind. By contrast, production activity has relocated evenly across both Services sectors at the expense of Goods: RI-Services and NRI-Services see similar expansions in their value added shares.

We proceed by developing a multi-sector growth model where the evolution of sectoral production and innovation activity together with productivity growth are jointly determined. In the model, the final-good produce accesses a non-homothetic technology with complementarities in sectoral goods. That way, the standard forces of structural change are present. On top of this, we add a directed technical change block à la [Acemoglu \(2002\)](#) whereby innovators develop sector-specific intermediate input varieties. In the model, the allocation of innovation activity is determined by the standard market size and relative price effects together with markups, sector-specific research productivity and knowledge spillovers. Innovation expands sectoral stocks of varieties and raises productivity, thereby feeding back into relative prices and expenditure patterns and, ultimately, the sectoral allocation of production and innovation.

We then calibrate our model to mimic structural change in both innovation and production activity as well as the sectoral productivity growth paths. Using the calibrated version of the model, we run a total of five quantitative exercises of which the first two are decompositions along the baseline path. In the first exercise, we decompose sectoral

productivity growth into endogenous innovation, embodied technical progress, relative price effects and labor efficiency. The decomposition reveals that endogenous innovation is the main source of sectoral productivity growth differences. This motivates our second exercises which decomposes endogenous innovation into research productivity, R&D employment and spillovers. We find that the Goods sector benefits from both improving research productivity, research employment and within-sector spillovers. RI-Services, in contrast, benefit from both cross-sector spillovers as well as improving research productivity and the reallocation of R&D workers while NRI-Services mostly from only cross-sector spillovers that offset declining research productivity.

We proceed with two general equilibrium counterfactual exercises. The third exercise quantifies assesses the role of income effects, relative prices and the markup drift in shaping aggregate productivity growth. We find that structural change has indeed shaped sectoral productivity paths, with relative prices playing a comparatively larger role than income effects and the markup drift. In the full shutdown case, we find that productivity growth would have been 2.13%, higher than both the fixed-shares benchmark and the baseline rates of 1.90% and 1.77%. The fourth exercise explores how heterogeneous innovation technologies and spillovers have affected paths of sectoral productivity. Innovation technologies differ by the 1950 research productivity, the subsequent research productivity growth rates and spillovers. Homogenizing research productivity growth rates shifts innovation activity towards NRI-Services, whereas homogenizing knowledge spillovers shift it strongly towards Goods.

The final exercise explicitly implements an exact Törnqvist decomposition into within-sector labor productivity, labor reallocation and market drift. This allows us to quantify the aggregate productivity losses from structural change into a composition component and an endogenous sectoral-productivity response. While the compositional effects of structural change have slowed down aggregate productivity, the endogenous sectoral productivity channel accounts for 62 percent of the slowdown in aggregate productivity growth relative to the world with no structural change.

Appendices

A Harmonized Industry Classification and NAICS Crosswalk

Our empirical analysis combines several data sources that use different industry classifications, including the industry-level productivity panel, IPUMS data on R&D workers, and patent data used to measure innovation and knowledge spillovers. We harmonize these sources using NAICS codes and map them into a common 44-industry classification covering 1947–2023.³⁰ Appendices B, C, and E describe the source-specific mappings into this common classification.

Table A.1 reports the resulting crosswalk and defines the three-sector aggregation used throughout the paper. We first divide industries into Goods and Services. We then separate Services according to research intensity, as described in Appendix E. RI-Services consists of Information (NAICS 51), Finance and Insurance (NAICS 52), and Professional, Scientific, and Technical Services (NAICS 54). All remaining service industries are classified as NRI-Services. The resulting Goods, RI-Services, and NRI-Services classification is used consistently across the productivity, employment, and innovation data and in the quantitative model.

³⁰We construct the industry-level productivity panel used in the paper; see [Elbadri & Vizcaino \(2025\)](#). The data and construction files are publicly available at [this repository](#). The panel is available in a 44-industry version covering 1947–2023 and a more disaggregated 63-industry version covering 1963–2023. We use the 44-industry version to retain the longer postwar sample.

Table A.1. Harmonized Industry Classification and NAICS Crosswalk

Broad sector	Industry	NAICS code	Model sector
Goods	Farms	11	Goods
	Forestry, fishing, and related activities	113–115	Goods
	Oil and gas extraction	211	Goods
	Mining, except oil and gas	212	Goods
	Support activities for mining	213	Goods
	Utilities	22	Goods
	Construction	23	Goods
	Food and beverage and tobacco products	311–312	Goods
	Textile mills and textile product mills	313–314	Goods
	Apparel and leather and allied products	315–316	Goods
	Wood products	321	Goods
	Paper products	322	Goods
	Printing and related support activities	323	Goods
	Petroleum and coal products	324	Goods
	Chemical products	325	Goods
	Plastics and rubber products	326	Goods
	Nonmetallic mineral products	327	Goods
	Primary metals	331	Goods
	Fabricated metal products	332	Goods
	Machinery	333	Goods
	Computer and electronic products	334	Goods
	Electrical equipment, appliances, and components	335	Goods
	Motor vehicles, bodies and trailers, and parts	3361–3363	Goods
	Other transportation equipment	3364–3369	Goods
	Furniture and related products	337	Goods
	Miscellaneous manufacturing	339	Goods
Services	Wholesale trade	42	NRI-Serv.
	Retail trade	44–45	NRI-Serv.
	Transportation and warehousing	48–49	NRI-Serv.
	Information	51	RI-Serv.
	Finance and insurance	52	RI-Serv.
	Real estate	531	NRI-Serv.
	Rental and leasing services and lessors of intangible assets	532–533	NRI-Serv.
	Professional, scientific, and technical services	54	RI-Serv.
	Management of companies and enterprises	55	NRI-Serv.
	Administrative and waste management services	56	NRI-Serv.
	Educational services	61	NRI-Serv.
	Health care and social assistance	62	NRI-Serv.
	Arts, entertainment, and recreation	71	NRI-Serv.
	Accommodation	721	NRI-Serv.
	Food services and drinking places	722	NRI-Serv.
	Other services, except government	81	NRI-Serv.
	Federal government	92	NRI-Serv.
	State and local government		NRI-Serv.

Notes: The table reports the 44-industry classification used to harmonize the paper’s empirical data sources. The first column divides industries into Goods and Services following the structural-change classification in [Herrendorf et al. \(2014\)](#). The second and third columns report industry names and corresponding NAICS codes. The final column maps each industry into one of the three sectors used in the analysis: Goods, research-intensive services (RI-Serv.), or non-research-intensive services (NRI-Serv.). RI-Serv. comprises Information, Finance and Insurance, and Professional, Scientific, and Technical Services; all remaining service industries are classified as NRI-Serv. Appendix E describes the research-intensity measure underlying this classification. Federal government and state and local government are reported separately in the underlying industry panel, although both fall under NAICS code 92.

B Identifying R&D Workers in IPUMS USA

To measure the sectoral allocation of R&D workers over time, we use IPUMS USA microdata from the 1950 (1%), 1960 (1%), 1970 (Form 1 State), 1980 (5%), 1990 (5%), and 2000 (5%) censuses and the 2010, 2020, and 2023 American Community Surveys (Ruggles et al. 2025). Because IPUMS does not directly identify whether an individual works in R&D, we construct an R&D-worker indicator from occupational classifications and harmonize both occupations and industries across years. We then aggregate the resulting microdata into the common industry classification described in Appendix A.

The construction proceeds in four steps:

1. We begin with the O*NET list of STEM occupations and retain occupations classified as “Research, Development, Design and Practitioners” or “Research, Development, Design and Practitioners; Technologists and Technicians.” We further restrict this set to occupations that we view as directly relevant for the production or application of knowledge and technology.³¹
2. We map the resulting O*NET-SOC occupation codes into the *OCCSOC* classification used by IPUMS using the IPUMS occupation crosswalk. This produces a list of R&D occupations expressed in *OCCSOC* codes.
3. Because *OCCSOC* is unavailable in the pre-2000 censuses, we construct an additional crosswalk from *OCCSOC* to *OCC1990*. We use the 2023 ACS, in which individuals are coded under both classifications, to map the R&D occupations identified above into *OCC1990*. We then use these codes to identify R&D workers consistently in the earlier census samples.
4. We similarly harmonize industries by constructing a crosswalk from *IND1990* to *INDNAICS* using post-1990 IPUMS observations for which both classifications are available. We then map individuals into the benchmark industries reported in Table A.1. Using the person weight *PERWT*, we aggregate R&D and non-R&D workers by industry and year to construct an industry-year panel of R&D employment and R&D intensity.³²

The crosswalks require some manual reconciliation where occupation or industry classifications do not map one-to-one across coding systems. We use these manual adjustments primarily to resolve missing or ambiguous links and to maintain consistent definitions across census years.

³¹Table B.1 reports all occupations in these two O*NET categories and indicates which are included in our R&D-worker measure.

³²We restrict the sample to employed individuals aged 21–65.

Table B.1. O*NET R&D Occupations and Inclusion in the R&D-Worker Measure

O*NET R&D occupation	O*NET-SOC code	Included?	O*NET R&D occupation	O*NET-SOC code	Included?
Actuaries	15-2011.00	✓	Landscape Architects	17-1012.00	✓
Acupuncturists	29-1291.00	×	Life Scientists, All Other	19-1099.00	✓
Acute Care Nurses	29-1141.01	×	Low Vision Therapists, Orientation and Mobility Specialists, and Vision Rehabilitation Therapists	29-1122.01	×
Advanced Practice Psychiatric Nurses	29-1141.02	×	Manufacturing Engineers	17-2112.03	✓
Aerospace Engineers	17-2011.00	✓	Marine Engineers and Naval Architects	17-2121.00	✓
Agricultural Engineers	17-2021.00	✓	Materials Engineers	17-2131.00	✓
Allergists and Immunologists	29-1229.01	×	Materials Scientists	19-2032.00	✓
Anesthesiologist Assistants	29-1071.01	×	Mathematical Science Occupations, All Other	15-2099.00	✓
Anesthesiologists	29-1211.00	×	Mathematicians	15-2021.00	✓
Animal Scientists	19-1011.00	✓	Mechanical Engineers	17-2141.00	✓
Anthropologists and Archeologists	19-3091.00	✓	Mechatronics Engineers	17-2199.05	✓
Architects, Except Landscape and Naval	17-1011.00	✓	Medical Scientists, Except Epidemiologists	19-1042.00	×
Art Therapists	29-1129.01	×	Microbiologists	19-1022.00	×
Astronomers	19-2011.00	✓	Microsystems Engineers	17-2199.06	✓
Athletic Trainers	29-9091.00	×	Midwives	29-9099.01	×
Atmospheric and Space Scientists	19-2021.00	✓	Mining and Geological Engineers, Including Mining Safety Engineers	17-2151.00	✓
Audiologists	29-1181.00	×	Molecular and Cellular Biologists	19-1029.02	×
Automotive Engineers	17-2141.02	✓	Music Therapists	29-1129.02	×
Biochemists and Biophysicists	19-1021.00	✓	Nanosystems Engineers	17-2199.09	✓
Bioengineers and Biomedical Engineers	17-2031.00	✓	Naturopathic Physicians	29-1299.01	×
Bioinformatics Scientists	19-1029.01	×	Network and Computer Systems Administrators	15-1244.00	✓
Bioinformatics Technicians	15-2099.01	✓	Neurologists	29-1217.00	×
Biological Scientists, All Other	19-1029.00	×	Neuropsychologists	19-3039.02	×
Biologists	19-1029.04	×	Nuclear Engineers	17-2161.00	✓
Biostatisticians	15-2041.01	✓	Nurse Anesthetists	29-1151.00	×
Blockchain Engineers	15-1299.07	✓	Nurse Midwives	29-1161.00	×
Business Intelligence Analysts	15-2051.01	✓	Nurse Practitioners	29-1171.00	×

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Table B.1. O*NET R&D Occupations and Inclusion in the R&D-Worker Measure (*continued*)

O*NET R&D occupation	O*NET-SOC code	Included?	O*NET R&D occupation	O*NET-SOC code	Included?
Cardiologists	29-1212.00	×	Obstetricians and Gynecologists	29-1218.00	×
Chemical Engineers	17-2041.00	✓	Occupational Therapists	29-1122.00	×
Chemists	19-2031.00	×	Operations Research Analysts	15-2031.00	✓
Chiropractors	29-1011.00	×	Ophthalmologists, Except Pediatric	29-1241.00	×
Civil Engineers	17-2051.00	✓	Optometrists	29-1041.00	×
Climate Change Policy Analysts	19-2041.01	✓	Oral and Maxillofacial Surgeons	29-1022.00	×
Clinical Data Managers	15-2051.02	✓	Orthodontists	29-1023.00	×
Clinical Neuropsychologists	19-3039.03	×	Orthopedic Surgeons, Except Pediatric	29-1242.00	×
Clinical Nurse Specialists	29-1141.04	×	Orthoptists	29-1299.02	×
Clinical and Counseling Psychologists	19-3033.00	×	Park Naturalists	19-1031.03	✓
Computer Hardware Engineers	17-2061.00	✓	Pediatric Surgeons	29-1243.00	×
Computer Network Architects	15-1241.00	✓	Pediatricians, General	29-1221.00	×
Computer Occupations, All Other	15-1299.00	✓	Penetration Testers	15-1299.04	✓
Computer Programmers	15-1251.00	✓	Petroleum Engineers	17-2171.00	✓
Computer Systems Analysts	15-1211.00	✓	Pharmacists	29-1051.00	×
Computer Systems Engineers/Architects	15-1299.08	✓	Photonics Engineers	17-2199.07	✓
Computer and Information Research Scientists	15-1221.00	✓	Physical Medicine and Rehabilitation Physicians	29-1229.04	×
Conservation Scientists	19-1031.00	✓	Physical Scientists, All Other	19-2099.00	×
Critical Care Nurses	29-1141.03	×	Physical Therapists	29-1123.00	×
Data Scientists	15-2051.00	✓	Physician Assistants	29-1071.00	×
Data Warehousing Specialists	15-1243.01	✓	Physicians, All Other	29-1229.00	×
Database Administrators	15-1242.00	✓	Physicians, Pathologists	29-1222.00	×
Database Architects	15-1243.00	✓	Physicists	19-2012.00	✓
Dental Hygienists	29-1292.00	×	Podiatrists	29-1081.00	×
Dentists, All Other Specialists	29-1029.00	×	Political Scientists	19-3094.00	✓
Dentists, General	29-1021.00	×	Preventive Medicine Physicians	29-1229.05	×
Dermatologists	29-1213.00	×	Prosthodontists	29-1024.00	×
Dietitians and Nutritionists	29-1031.00	×	Psychiatrists	29-1223.00	×

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Table B.1. O*NET R&D Occupations and Inclusion in the R&D-Worker Measure (*continued*)

O*NET R&D occupation	O*NET-SOC code	Included?	O*NET R&D occupation	O*NET-SOC code	Included?
Digital Forensics Analysts	15-1299.06	✓	Psychologists, All Other	19-3039.00	×
Document Management Specialists	15-1299.03	✓	Radiation Therapists	29-1124.00	×
Economists	19-3011.00	✓	Radio Frequency Identification Device Specialists	17-2072.01	✓
Electrical Engineers	17-2071.00	✓	Radiologists	29-1224.00	×
Electronics Engineers, Except Computer	17-2072.00	✓	Range Managers	19-1031.02	✓
Emergency Medicine Physicians	29-1214.00	×	Recreational Therapists	29-1125.00	×
Energy Engineers, Except Wind and Solar	17-2199.03	✓	Registered Nurses	29-1141.00	×
Engineers, All Other	17-2199.00	✓	Remote Sensing Scientists and Technologists	19-2099.01	×
Environmental Economists	19-3011.01	✓	Respiratory Therapists	29-1126.00	×
Environmental Engineers	17-2081.00	✓	Robotics Engineers	17-2199.08	✓
Environmental Restoration Planners	19-2041.02	✓	School Psychologists	19-3034.00	×
Environmental Scientists and Specialists, Including Health	19-2041.00	✓	Social Scientists and Related Workers, All Other	19-3099.00	✓
Epidemiologists	19-1041.00	×	Sociologists	19-3041.00	✓
Exercise Physiologists	29-1128.00	×	Software Developers	15-1252.00	✓
Family Medicine Physicians	29-1215.00	×	Software Quality Assurance Analysts and Testers	15-1253.00	✓
Fire-Prevention and Protection Engineers	17-2111.02	×	Soil and Plant Scientists	19-1013.00	✓
Food Scientists and Technologists	19-1012.00	✓	Solar Energy Systems Engineers	17-2199.11	✓
Foresters	19-1032.00	✓	Speech-Language Pathologists	29-1127.00	×
Fuel Cell Engineers	17-2141.01	✓	Sports Medicine Physicians	29-1229.06	×
General Internal Medicine Physicians	29-1216.00	×	Statisticians	15-2041.00	✓
Genetic Counselors	29-9092.00	×	Surgeons, All Other	29-1249.00	×
Geneticists	19-1029.03	×	Surgical Assistants	29-9093.00	×
Geographers	19-3092.00	✓	Survey Researchers	19-3022.00	✓
Geographic Information Systems Technologists and Technicians	15-1299.02	✓	Telecommunications Engineering Specialists	15-1241.01	✓
Geoscientists, Except Hydrologists and Geographers	19-2042.00	✓	Therapists, All Other	29-1129.00	×

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Table B.1. O*NET R&D Occupations and Inclusion in the R&D-Worker Measure (*continued*)

O*NET R&D occupation	O*NET-SOC code	Included?	O*NET R&D occupation	O*NET-SOC code	Included?
Health Informatics Specialists	15-1211.01	✓	Transportation Engineers	17-2051.01	✓
Health Information Technologists and Medical Registrars	29-9021.00	×	Transportation Planners	19-3099.01	✓
Health and Safety Engineers, Except Mining Safety Engineers and Inspectors	17-2111.00	×	Urban and Regional Planners	19-3051.00	✓
Healthcare Diagnosing or Treating Practitioners, All Other	29-1299.00	×	Urologists	29-1229.03	×
Healthcare Practitioners and Technical Workers, All Other	29-9099.00	×	Validation Engineers	17-2112.02	✓
Hospitalists	29-1229.02	×	Veterinarians	29-1131.00	×
Human Factors Engineers and Ergonomists	17-2112.01	✓	Video Game Designers	15-1255.01	✓
Hydrologists	19-2043.00	✓	Water/Wastewater Engineers	17-2051.02	✓
Industrial Ecologists	19-2041.03	✓	Web Administrators	15-1299.01	✓
Industrial Engineers	17-2112.00	✓	Web Developers	15-1254.00	✓
Industrial-Organizational Psychologists	19-3032.00	✓	Web and Digital Interface Designers	15-1255.00	✓
Information Security Analysts	15-1212.00	✓	Wind Energy Engineers	17-2199.10	✓
Information Security Engineers	15-1299.05	✓	Zoologists and Wildlife Biologists	19-1023.00	✓
Information Technology Project Managers	15-1299.09	✓			

Notes: We use “O*NET R&D occupations” to refer to occupations within O*NET’s STEM occupation list that are classified as either “Research, Development, Design and Practitioners” or “Research, Development, Design and Practitioners; Technologists and Technicians.” The first and fourth columns report occupation titles, while the second and fifth columns report the corresponding O*NET-SOC codes. The third and sixth columns indicate whether each occupation is included in our measure of R&D workers. A check mark indicates inclusion and a cross indicates exclusion.

C Mapping Patents to Industries and Broad Sectors

To map patents into the industry classification used in the paper, we combine the U.S. Patent and Trademark Office Historical Patent Data Files with the probabilistic USPC–NAICS concordance constructed by [Goldschlag et al. \(2016\)](#).³³ We use the version that maps U.S. Patent Classification (USPC) codes into the 2007 NAICS classification and includes both manufacturing and services. Because the industries in Table A.1 are defined at different levels of NAICS aggregation, we use both the two-digit and three-digit versions of the concordance.³⁴

The construction proceeds in four steps:

1. We restrict the patent data to patents that are ultimately granted and assign each patent to its application year using the filing date. We then count patents by USPC class and application year.
2. We merge these counts with the probabilistic USPC–NAICS concordance. Because a given USPC class can map into several NAICS industries, each USPC–NAICS pair is associated with a probability, with probabilities summing to one within USPC class.
3. For each USPC–NAICS-year observation, we multiply the number of patents in the USPC class by the corresponding concordance probability. This yields the number of patents probabilistically allocated to a given NAICS industry from that USPC class. We then aggregate across USPC classes to obtain patent counts by NAICS industry and year. We perform this procedure separately using the two-digit and three-digit concordances.
4. We combine the resulting datasets, retain the NAICS industries appearing in Table A.1, and map them into Goods, RI-Services, and NRI-Services using the harmonized classification described in Appendix A. Aggregating across industries then yields annual patent counts for each of the three model sectors.

³³The concordance is publicly available at [this website](#).

³⁴Some manufacturing industries in Table A.1 are defined using four-digit NAICS codes. Because all manufacturing industries are assigned to Goods, finer disaggregation is unnecessary for constructing the three model sectors.

D Constructing Cross-Sector Citation Probabilities

This appendix describes the construction of the cross-sector citation probabilities plotted in Figure 3, which we use to discipline the knowledge-spillover parameters $\delta_{j\ell}$. The construction combines patent citations from the OECD Citations database, patent technology classifications from PatentsView, and a probabilistic concordance between patent technologies and industries.

Data

Patent citations come from the February 2025 edition of the OECD Citations database, derived from the Autumn 2024 edition of the EPO Worldwide Patent Statistical Database (PATSTAT) (Webb et al. 2005). We use citations made by patents granted by the U.S. Patent and Trademark Office (USPTO). Each citation links a citing patent to a cited patent application through its PATSTAT application identifier.

Technology classifications come from PatentsView,³⁵ which reports the International Patent Classification (IPC) codes assigned to U.S. patents at issuance. We use IPC codes at the four-character level, combining the IPC section, class, and subclass. We refer to classifications at this level as IPC4 codes. This is also the level at which we link patent technologies to industries.

To map technologies into industries, we use the Algorithmic Links with Probabilities concordance constructed by Goldschlag et al. (2016). The concordance assigns each IPC4 code probabilistically across ISIC Rev. 3.1 industries, with the industry weights for each IPC4 code summing to one. We use the four-digit version of the concordance and subsequently map the resulting ISIC industries into the Goods, RI-Services, and NRI-Services sectors used in the paper, as described below.

The benchmark sample uses citing patents granted between 1976 and 2020. The starting year is determined by the availability of technology classifications in PatentsView.

Allocating Citations across Technologies

Both the citing and cited sides of a citation can map to multiple technologies. A patent may carry several IPC4 codes, while a cited application may correspond to several U.S. patent publications. We therefore use fractional weights so that each citation contributes a total weight of one.

Let $\mathcal{I}_p$ denote the set of distinct IPC4 codes assigned to citing patent p . We distribute

³⁵PatentsView is available at patentsview.org.

the patent's weight equally across these technologies:

$$\omega_p(i) = \frac{\mathbf{1}\{i \in \mathcal{I}_p\}}{|\mathcal{I}_p|}, \quad \sum_i \omega_p(i) = 1.$$

Cited applications are identified in the OECD data by their PATSTAT application identifiers. We link each cited application a to its granted U.S. equivalents using the equivalents file supplied with the citations database. Let $\mathcal{E}_a$ denote the set of granted U.S. equivalents associated with application a . We define the technology distribution of the cited application as the average of the technology distributions of these equivalents:

$$\omega_a(j) = \frac{1}{|\mathcal{E}_a|} \sum_{p \in \mathcal{E}_a} \omega_p(j), \quad \sum_j \omega_a(j) = 1.$$

For a citation from patent p to application a , we assign weight $\omega_p(i)\omega_a(j)$ to the ordered pair consisting of citing technology i and cited technology j . These weights sum to one across all technology pairs associated with the citation. Aggregating across citations yields the technology-level citation counts

$$C_{ij} = \sum_{(p,a)} \omega_p(i)\omega_a(j), \tag{D.1}$$

where C_{ij} measures the citation weight running from citing technology i to cited technology j . By construction, $\sum_{i,j} C_{ij}$ equals the number of citations in the matched sample.

Aggregating Technologies to Sectors

We next map technology-level citation weights into the three sectors used in the paper. Let A_{ik} denote the probability weight that the concordance assigns to industry k for technology i , with

$$\sum_k A_{ik} = 1.$$

Let $\mathcal{S}(k) \in \{g, rs, ns\}$ denote the model sector to which industry k belongs, as defined in Appendix A. We aggregate the industry weights associated with technology i to sector s according to

$$S_{is} = \sum_{k: \mathcal{S}(k)=s} A_{ik}, \quad \sum_s S_{is} = 1. \tag{D.2}$$

Thus, S_{is} is the share of technology i that the concordance assigns to sector s .

For each year t , let C_t denote the technology-level citation counts constructed above using

citations made by patents granted in year t . Sector-level citation counts are then

$$M_t = S' C_t S. \quad (\text{D.3})$$

The entry $M_{j\ell,t}$ is the citation weight running from patents in citing sector j to patents associated with cited sector ℓ . Because the rows of S sum to one, aggregation preserves the total citation weight in each year.

Finally, we normalize the citation weights of each citing sector to sum to one:

$$\delta_{j\ell}(t) = \frac{M_{j\ell,t}}{\sum_{\ell'} M_{j\ell',t}}, \quad \sum_{\ell} \delta_{j\ell}(t) = 1. \quad (\text{D.4})$$

Hence, $\delta_{j\ell}(t)$ is the probability that a patent in sector j cites knowledge associated with sector ℓ , conditional on making a citation. These are the cross-sector citation probabilities plotted in Figure 3 and used to discipline the knowledge-spillover weights in the model.

Mapping ISIC Industries to Model Sectors

The patent-industry concordance is defined in ISIC Rev. 3.1, whereas the three-sector classification used elsewhere in the paper is defined using NAICS codes. We therefore map ISIC industries into the Goods, RI-Services, and NRI-Services sectors defined in Appendix A.

Several ISIC Rev. 3.1 divisions contain activities that belong to different model sectors under our NAICS-based classification. We therefore assign these industries at the four-digit ISIC class level rather than assigning the entire division to a single sector. Table D.1 reports the classes for which this finer assignment matters. The complete ISIC-to-model-sector concordance is provided in the replication package.

Robustness

Table D.2 reports cross-sector citation probabilities under several alternative samples. The first three alternative specifications split the benchmark sample by the grant year of the citing patent. We then distinguish between citations introduced by patent examiners and those submitted by applicants. Finally, we restrict attention to citations classified as category X or Y, which identify prior art that is particularly relevant for assessing novelty either on its own or in combination with other documents.

The citation probabilities are broadly similar when we distinguish between examiner-added and applicant-submitted citations. The largest difference across the two samples is 6.3 percentage points, for the share of NRI-Services citations directed toward Goods, while the qualitative ordering of citation sources is unchanged across sectors. The benchmark

Table D.1. ISIC Rev. 3.1 Classes Requiring Separate Sector Assignment

ISIC class	Description	NAICS code	Model sector
2211–2219	Publishing, including software	511	RI-Serv.
2221–2230	Printing and reproduction of media	323, 334	Goods
6411–6412	Post and courier activities	491, 492	NRI-Serv.
6420	Telecommunications	517	RI-Serv.
7250	Repair of office and computing machinery	811	NRI-Serv.
7491–7499	Labour recruitment, security, cleaning, and packaging	561	NRI-Serv.
9211–9213	Motion picture, radio, and television activities	512, 515	RI-Serv.
9220	News agency activities	519	RI-Serv.
9214–9219	Live entertainment	711	NRI-Serv.
9231–9249	Libraries, museums, sports, and recreation	712, 713	NRI-Serv.

Notes: The table reports ISIC Rev. 3.1 classes that are assigned separately rather than according to their broader ISIC division. Assigning entire divisions would classify software publishing as Goods, place telecommunications and broadcasting outside RI-Services, and classify business-support activities as RI-Services, contrary to the NAICS-based sector definitions used in the analysis. The third column reports the corresponding NAICS codes, and the final column reports the assignment to Goods, research-intensive services (RI-Serv.), or non-research-intensive services (NRI-Serv.). Appendix A reports the benchmark industry classification, and Appendix E describes the research-intensity criterion separating RI-Services from NRI-Services.

Table D.2. Cross-Sector Citation Probabilities: Alternative Samples

Parameter	Benchmark	1976–1990	1991–2005	2006–2020	Examiner	Applicant	X or Y
$\delta_{m,m}$	0.910	0.976	0.952	0.903	0.913	0.909	0.862
$\delta_{m,rs}$	0.055	0.011	0.029	0.060	0.063	0.052	0.101
$\delta_{m,ns}$	0.035	0.013	0.019	0.038	0.024	0.039	0.037
$\delta_{rs,m}$	0.685	0.827	0.793	0.678	0.661	0.695	0.699
$\delta_{rs,rs}$	0.257	0.160	0.179	0.262	0.293	0.241	0.245
$\delta_{rs,ns}$	0.058	0.013	0.028	0.060	0.046	0.064	0.056
$\delta_{ns,m}$	0.736	0.798	0.780	0.732	0.685	0.748	0.712
$\delta_{ns,rs}$	0.107	0.011	0.046	0.112	0.127	0.102	0.149
$\delta_{ns,ns}$	0.157	0.191	0.174	0.156	0.188	0.150	0.140

Notes: The table reports cross-sector citation probabilities for the benchmark sample and alternative subsamples. The parameter $\delta_{j\ell}$ denotes the probability that a patent in citing sector j cites knowledge associated with sector ℓ , conditional on making a citation. For each citing sector j , the probabilities $\delta_{j,m}$, $\delta_{j,rs}$, and $\delta_{j,ns}$ therefore sum to one by construction. The sectors are Goods (m), research-intensive services (rs), and non-research-intensive services (ns). The 1976–1990, 1991–2005, and 2006–2020 columns restrict the sample according to the grant year of the citing patent. The Examiner and Applicant columns restrict the sample to examiner-added and applicant-submitted citations, respectively. The final column restricts the sample to citations classified as category X or Y.

pattern is therefore not driven by a particular citation origin.

Restricting the sample to category X and Y citations also preserves the main structure of cross-sector knowledge flows. Goods remains the dominant source of cited knowledge for all three sectors, while RI-Services and NRI-Services continue to place substantially more weight on external knowledge than Goods does.

E Classifying Services by Research Intensity

Appendix B describes how we construct industry-level measures of R&D employment from IPUMS USA. We use these measures to divide service industries into research-intensive services (RI-Services) and non-research-intensive services (NRI-Services). The Goods–Services distinction follows the harmonized industry classification in Appendix A; the purpose of this appendix is to explain the additional split within Services.

We define an industry’s research intensity as the ratio of R&D workers to non-R&D workers. Table E.1 reports this measure by census year, together with its average over 1970–2020 and the resulting model-sector classification. We base the classification on the 1970–2020 average so that it reflects persistent differences in research intensity rather than any single-year observation. We exclude 1950 and 1960 from the benchmark average because several service industries are not separately identified in those years; including them for industries with available observations does not materially affect the classification.

We use Manufacturing’s average research intensity over 1970–2020 as the threshold. Service industries with an unrounded average research intensity above this threshold are classified as RI-Services, while the remaining service industries are classified as NRI-Services.

Table E.1. R&D Intensity by Service Industry

Industry	1950	1960	1970	1980	1990	2000	2010	2020	2023	1970–2020 avg.	Model sector
Information	0.07	0.06	0.09	0.12	0.17	0.37	0.40	0.63	0.65	0.29	RI-Serv.
Professional, scientific, and technical services	0.08	0.16	0.26	0.25	0.22	0.28	0.26	0.31	0.33	0.26	RI-Serv.
Finance and insurance	0.02	0.01	0.04	0.06	0.08	0.11	0.12	0.18	0.21	0.10	RI-Serv.
<i>Manufacturing (threshold)</i>	0.03	0.04	0.07	0.07	0.09	0.10	0.11	0.15	0.16	0.10	Goods
Federal, state, and local government	0.05	0.05	0.09	0.08	0.08	0.09	0.09	0.12	0.13	0.09	NRI-Serv.
Federal, state, and local government	0.05	0.05	0.09	0.08	0.08	0.09	0.09	0.12	0.13	0.09	NRI-Serv.
Educational services	0.02	0.02	0.02	0.02	0.03	0.05	0.05	0.06	0.06	0.04	NRI-Serv.
Health care and social assistance	0.01	0.01	0.02	0.03	0.04	0.04	0.05	0.06	0.06	0.04	NRI-Serv.
Wholesale trade	0.01	0.01	0.02	0.02	0.04	0.03	0.03	0.05	0.04	0.03	NRI-Serv.
Arts, entertainment, and recreation	0.00	0.00	0.00	0.03	0.05	0.01	0.01	0.02	0.03	0.02	NRI-Serv.
Rental and leasing services and lessors of intangible assets	NA	NA	NA	NA	0.01	0.02	0.03	0.03	0.04	0.02	NRI-Serv.
Administrative and waste management services	NA	NA	0.02	0.03	0.02	0.01	0.01	0.02	0.02	0.02	NRI-Serv.
Retail trade	0.00	0.00	0.00	0.01	0.01	0.03	0.02	0.03	0.03	0.02	NRI-Serv.
Transportation and warehousing	0.00	0.00	0.01	0.01	0.01	0.01	0.02	0.02	0.02	0.01	NRI-Serv.
Real estate	0.01	0.00	0.01	0.01	0.01	0.01	0.01	0.02	0.03	0.01	NRI-Serv.
Other services, except government	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.02	0.02	0.01	NRI-Serv.
Accommodation	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.02	0.02	0.01	NRI-Serv.
Food services and drinking places	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	NRI-Serv.

Notes: The underlying data are from IPUMS USA (Ruggles et al. 2025). The table reports R&D intensity for service industries and Manufacturing, which serves as the classification threshold. R&D intensity is the ratio of the weighted number of R&D workers to the weighted number of non-R&D workers. Appendix B describes the occupation-based identification of R&D workers. The penultimate column reports average R&D intensity over 1970–2020. Service industries with an unrounded average above Manufacturing are classified as research-intensive services (RI-Serv.); all others are classified as non-research-intensive services (NRI-Serv.). Using unrounded averages explains why Finance and Insurance and Manufacturing display the same rounded value but receive different assignments. NA indicates that an industry is not separately identified in that year.

F Constructing Machine-Producer Markups

This appendix describes how we construct sectoral time series for machine-producer markups over 1947–2023. The quantitative model uses the resulting series over 1950–2023.

In the model, the price charged by a machine producer in sector j is a markup $1/\mu_j^x(t)$ over marginal cost, and the corresponding operating profit share of machine revenue is $1 - \mu_j^x(t)$. We construct an empirical counterpart to this profit share using the difference between capital rental prices based on industry-specific internal rates of return and rental prices based on an economy-wide external rate of return. After aggregating capital assets within an industry, define

$$\psi_i(t) = \frac{R_i^{IRR}(t) - R_i^{ERR}(t)}{R_i^{IRR}(t)}. \quad (\text{F.1})$$

We interpret $\psi_i(t)$ as the empirical counterpart to the machine-producer profit share and therefore set

$$1 - \mu_i^x(t) = \psi_i(t), \quad \frac{1}{\mu_i^x(t)} = \frac{1}{1 - \psi_i(t)}.$$

The remainder of the appendix describes the construction of $R_i^{IRR}(t)$ and $R_i^{ERR}(t)$ and their aggregation into the three model sectors.

Data

Our primary source is the Bureau of Labor Statistics (BLS) Rental Prices dataset,³⁶ which reports asset-by-industry rental prices constructed using both internal and external rates of return from 1987 onward (U.S. Bureau of Labor Statistics 2025b). To extend these series back to 1947, we follow the methodology described in the BLS Handbook of Methods (U.S. Bureau of Labor Statistics 2025a) and combine several additional data sources.

First, the Bureau of Economic Analysis (BEA) Detailed Fixed Assets dataset provides asset-by-industry measures of nominal capital stocks, real capital stocks, and real investment beginning in 1947 (U.S. Bureau of Economic Analysis 2004). We restrict attention to private nonresidential fixed assets and exclude consumer durables. These data allow us to construct asset prices, depreciation rates, and capital gains.

Second, we obtain industry-level capital compensation from U.S. KLEMS for 1947–2014 (Jorgenson et al. 2017). We extend the capital-compensation series beyond 2014 using the industry-level data in Elbadri & Vizcaino (2025), linking the two sources at their overlap.

³⁶The BLS Rental Prices tables are available at [this website](#).

Finally, we use Penn World Table 11.0 to obtain an economy-wide rate of return for the United States (Feenstra et al. 2015). We use this series as the external rate of return when constructing rental prices before they become available directly from the BLS.

Industry Harmonization

The construction is carried out at the industry level before aggregation into Goods, RI-Services, and NRI-Services. The working classification contains 61 private industries. The underlying datasets are generally available at this level, although the BEA Detailed Fixed Assets data are more disaggregated for some industries. We aggregate these finer BEA categories to the common industry classification before combining the datasets.

For nominal quantities, such as capital stocks or compensation, aggregation is by summation. When calculations require averaging across asset-industry observations, we first combine the finer industry categories into their corresponding common industry group and then apply the weighting procedure described below.

Because the BEA fixed-asset data used in this construction cover private nonresidential assets, the markup series for RI-Services and NRI-Services are constructed using private service industries only. We use these private-sector markup series as the empirical counterparts to the corresponding service-sector markups in the model.

Construction

Let k index capital assets, i industries, and t years. The construction proceeds in five steps.

1. Construct asset prices, depreciation, and capital gains.

We combine nominal capital stocks, real capital stocks in 2017 prices, and real investment from the BEA Detailed Fixed Assets dataset. Let $\mathcal{V}_{ki}(t)$ denote the nominal value of the capital stock, $K_{ki}(t)$ the corresponding real stock, and $I_{ki}(t)$ real investment. The asset price deflator is

$$P_{ki}(t) = \frac{\mathcal{V}_{ki}(t)}{K_{ki}(t)}.$$

We construct the wealth-stock depreciation rate as

$$\delta_{ki}(t) = \frac{I_{ki}(t) - [K_{ki}(t) - K_{ki}(t-1)]}{[K_{ki}(t) + K_{ki}(t-1)]/2}.$$

Finally, we construct the capital-gain term from a three-year moving average of the

asset price:

$$\dot{P}_{ki}(t) = \tilde{P}_{ki}(t) - \tilde{P}_{ki}(t-1), \quad \tilde{P}_{ki}(t) = \frac{P_{ki}(t-1) + P_{ki}(t) + P_{ki}(t+1)}{3}.$$

2. Construct industry-level internal and external rates of return.

The BLS rental-price methodology requires an industry-specific internal rate of return and an economy-wide external rate of return. Prior to 1987, we construct the industry-specific internal rate of return following the BLS Handbook of Methods:

$$IRR_i(t) = \frac{CC_i(t) - OC_i(t)}{SV_i(t)},$$

where $CC_i(t)$ denotes industry capital compensation, $OC_i(t)$ ownership costs, and $SV_i(t)$ the value of the industry's capital stock.

Abstracting from taxes, ownership costs and stock value are

$$OC_i(t) = \sum_k K_{ki}(t) \left[P_{ki}(t) \delta_{ki}(t) - \dot{P}_{ki}(t) \right]$$

and

$$SV_i(t) = \sum_k K_{ki}(t) P_{ki}(t).$$

Capital compensation for this pre-1987 construction comes from U.S. KLEMS, while the ownership-cost and stock-value terms are computed from the asset-level series constructed in step 1.

From 1987 onward, industry-level internal rates of return are available directly from the BLS Rental Prices dataset. We use the internally constructed series to recover pre-1987 variation and anchor its level to the BLS series in 1987:

$$IRR_i(t) = \begin{cases} IRR_i^{BLS}(t), & t \geq 1987, \\ IRR_i^{BLS}(1987) \left(\frac{IRR_i^{CON}(t)}{IRR_i^{CON}(1987)} \right), & t < 1987, \end{cases}$$

where $IRR_i^{CON}(t)$ denotes the internal rate of return constructed from the preceding data sources.

We use the Penn World Table economy-wide U.S. rate of return as the external rate of return, denoted $ERR(t)$. Following the BLS methodology, when the constructed industry-specific internal rate of return is negative, we replace it with the external rate of return.

3. Construct rental prices using the internal rate of return.

Given $IRR_i(t)$, we construct the internal-rate rental price for asset k in industry i as

$$R_{ki}^{IRR}(t) = IRR_i(t)P_{ki}(t-1) + P_{ki}(t)\delta_{ki}(t) - \dot{P}_{ki}(t).$$

For 1947–1986, we compute this expression using the asset-level data from step 1 and the internal rates of return from step 2. From 1987 onward, the corresponding asset-by-industry rental prices are available directly from the BLS Rental Prices dataset.

We then aggregate rental prices across asset types within each industry. The weights are estimated asset-level capital compensation,

$$R_{ki}^{IRR}(t)K_{ki}(t).$$

If this quantity is negative for a given asset-industry observation, we instead use

$$R_{ki}^{ERR}(t)K_{ki}(t)$$

as the weight.

This yields an industry-level internal-rate rental-price series constructed from alternative sources for 1947–1986 and an industry-level series obtained from the BLS data from 1987 onward. We link their levels at 1987:

$$R_i^{IRR}(t) = \begin{cases} R_i^{IRR,BLS}(t), & t \geq 1987, \\ R_i^{IRR,BLS}(1987) \left(\frac{R_i^{IRR,CON}(t)}{R_i^{IRR,CON}(1987)} \right), & t < 1987. \end{cases}$$

4. Construct rental prices using the external rate of return.

We repeat the preceding procedure using the economy-wide external rate of return. At the asset-industry level, the external-rate rental price is

$$R_{ki}^{ERR}(t) = ERR(t)P_{ki}(t-1) + P_{ki}(t)\delta_{ki}(t).$$

Following the BLS methodology, the external-rate rental price excludes the capital-gain term that enters the internal-rate construction.

We aggregate these rental prices across assets using the same capital-compensation weights as in step 3. That is, we use $R_{ki}^{IRR}(t)K_{ki}(t)$ whenever it is positive and $R_{ki}^{ERR}(t)K_{ki}(t)$ otherwise. We again link the pre-1987 constructed series to the BLS

series at the 1987 level. Steps 3 and 4 therefore produce industry-level internal- and external-rate rental-price series over 1947–2023.

5. Construct profit shares and aggregate to model sectors.

We use the difference between the two rental-price measures to construct the empirical machine-producer profit share:

$$\psi_i(t) = 1 - \mu_i^x(t) = \frac{R_i^{IRR}(t) - R_i^{ERR}(t)}{R_i^{IRR}(t)}.$$

To limit the influence of measurement error and rule out values that are inadmissible in the model, we bound industry-level profit shares to the interval $[0.02, 0.98]$ before aggregating them to the three model sectors.

Sectoral profit shares are capital-compensation-weighted averages of the corresponding industry-level values. We obtain industry-level capital compensation from U.S. KLEMS through 2014 and extend the series through 2023 using [Elbadri & Vizcaino \(2025\)](#). We link the two sources at their 2014 overlap:

$$CC_i(t) = \begin{cases} CC_i^{KLEMS}(t), & t \leq 2014, \\ CC_i^{KLEMS}(2014) \left(\frac{CC_i^{EV}(t)}{CC_i^{EV}(2014)} \right), & t > 2014, \end{cases}$$

where $CC_i^{KLEMS}(t)$ denotes capital compensation from U.S. KLEMS and $CC_i^{EV}(t)$ denotes capital compensation from [Elbadri & Vizcaino \(2025\)](#).

For each model sector j , we then compute the capital-compensation-weighted profit share

$$1 - \mu_j^x(t) = \frac{\sum_{i \in j} CC_i(t) [1 - \mu_i^x(t)]}{\sum_{i \in j} CC_i(t)},$$

and obtain the corresponding machine-producer markup as

$$\frac{1}{\mu_j^x(t)}.$$

For RI-Services and NRI-Services, these sectoral averages use private service industries only, consistent with the coverage of the fixed-asset data used in the construction.

G Constructing Relative Capital Prices

This appendix describes how we construct relative capital-price indices for Goods, RI-Services, and NRI-Services over 1947–2023. The quantitative model uses the resulting series over 1950–2023. We use the same underlying asset-level data, industry classification, and rental-price measures as in Appendix F.

The construction proceeds in five steps.

1. Construct asset-level capital-price growth.

Let $P_{ki}(t)$ denote the price deflator for capital asset k in industry i in year t . From 1987 onward, these deflators are available directly from the BLS Rental Prices dataset. We compute annual log price growth as

$$g_{ki}^P(t) = \ln P_{ki}(t) - \ln P_{ki}(t-1).$$

For the period before 1987, we construct the asset price deflator from the BEA Detailed Fixed Assets data as

$$P_{ki}(t) = \frac{\mathcal{V}_{ki}(t)}{K_{ki}(t)},$$

where $\mathcal{V}_{ki}(t)$ is the nominal value of the capital stock and $K_{ki}(t)$ is the corresponding real stock in 2017 prices. We then compute $g_{ki}^P(t)$ in the same way. Combining the two sources provides asset-industry capital-price growth rates over 1948–2023.

2. Aggregate asset-level price growth to industries.

We aggregate capital-price growth across assets within each industry using Törnqvist weights based on estimated capital compensation. Define

$$R_{ki}^X(t) = \begin{cases} R_{ki}^{IRR}(t), & R_{ki}^{IRR}(t) > 0, \\ R_{ki}^{ERR}(t), & R_{ki}^{IRR}(t) \leq 0, \end{cases}$$

where $R_{ki}^{IRR}(t)$ and $R_{ki}^{ERR}(t)$ are the internal- and external-rate rental prices constructed in Appendix F. The asset's share of industry capital compensation is

$$s_{ki}(t) = \frac{R_{ki}^X(t)K_{ki}(t)}{\sum_{k'} R_{k'i}^X(t)K_{k'i}(t)},$$

and its Törnqvist weight between $t-1$ and t is

$$\bar{s}_{ki}(t) = \frac{s_{ki}(t-1) + s_{ki}(t)}{2}.$$

Industry-level capital-price growth is therefore

$$g_i^P(t) = \sum_k \bar{s}_{ki}(t) g_{ki}^P(t).$$

For the pre-1987 period, the rental-price measures entering these weights are constructed as described in Appendix F; from 1987 onward, the corresponding BLS measures are used. Chaining the resulting growth rates yields a consistent industry-level capital-price series over the full sample.

3. Aggregate industry-level price growth to model sectors.

We next aggregate industry-level capital-price growth into Goods, RI-Services, and NRI-Services using industry capital compensation as weights. As in Appendix F, capital compensation is taken from U.S. KLEMS through 2014 and extended through 2023 using [Elbadri & Vizcaino \(2025\)](#), with the two series linked at their 2014 overlap.

For industry i belonging to model sector j , define its capital-compensation share as

$$s_i^K(t) = \frac{CC_i(t)}{\sum_{i' \in j} CC_{i'}(t)},$$

with Törnqvist weight

$$\bar{s}_i^K(t) = \frac{s_i^K(t-1) + s_i^K(t)}{2}.$$

Sectoral capital-price growth is then

$$g_j^{PK}(t) = \sum_{i \in j} \bar{s}_i^K(t) g_i^P(t), \quad j \in \{m, rs, ns\}.$$

Consistent with Appendix F, the RI-Services and NRI-Services indices are constructed using private service industries only.

4. Construct sectoral capital-price indices.

We convert the sectoral log growth rates into chained price indices normalized to one in 1947:

$$P_j^K(t) = \exp \left(\sum_{\tau=1948}^t g_j^{PK}(\tau) \right), \quad P_j^K(1947) = 1.$$

Thus, $P_j^K(t)$ measures the cumulative change in the price of the sector-specific capital bundle relative to its 1947 level.

5. Express capital prices relative to the aggregate value-added price.

The model requires the price of capital relative to the economy-wide price of output. We therefore deflate each sectoral capital-price index by an aggregate value-added price index.

We construct industry-level value-added price growth from the data in [Elbadri & Vizcaino \(2025\)](#). For each industry, we form a nominal value-added index with the same 1947 base year as the corresponding real value-added quantity index and divide the nominal index by the quantity index to obtain an industry-level value-added price index. We then compute annual log growth rates and aggregate them across industries using Törnqvist weights based on nominal value-added shares.

Let $P^{VA}(t)$ denote the resulting economy-wide value-added price index, normalized to one in 1947. The relative capital-price index for sector j is

$$\tilde{P}_j^K(t) = \frac{P_j^K(t)}{P^{VA}(t)}. \quad (\text{G.1})$$

These relative capital-price series are the empirical objects used in the calibration.

H Counterfactual Exercises

H.1 Subperiod Decomposition of Sectoral Labor Productivity Growth

Table 3 in the main text reports the decomposition of sectoral labor productivity growth over 1950–2023. This appendix reports the corresponding decomposition separately for 1950–1980 and 1980–2023. The subperiod results show that the importance of endogenous innovation for the productivity-growth differential is not driven by averaging across the full postwar period.

The subperiod decomposition reinforces the main result. In Goods, the contribution of endogenous innovation rises from 2.30 percentage points per year in 1950–1980 to 2.59 percentage points in 1980–2023, even as overall labor productivity growth falls from 2.61 to 2.37 percent. The slowdown instead reflects a more negative relative-price effect, whose contribution falls from -0.13 to -0.66 percentage points. RI-Services experiences stronger contributions from both endogenous innovation and embodied technical progress after 1980, while NRI-Services continues to exhibit substantially weaker innovation-driven productivity growth in both subperiods.

Table H.1. Decomposition of Sectoral Labor Productivity Growth, 1950–1980

	Sector		
	Goods	RI-Serv.	NRI-Serv.
I. Endogenous innovation	2.30	0.77	0.61
(a) <i>Idea growth</i>	2.31	0.77	0.61
(b) <i>Elasticity drift</i>	−0.01	0.00	0.00
II. Embodied technical progress	0.07	0.56	0.15
(a) <i>Machine productivity</i>	−0.06	0.62	0.10
(b) <i>Machine market power</i>	0.13	−0.06	0.06
III. Relative price effect	−0.13	0.37	0.37
(a) <i>Output price</i>	−0.18	0.18	0.14
(b) <i>Sectoral market power</i>	0.05	0.19	0.24
IV. Labor efficiency	0.38	0.41	0.39
Total	2.61	2.11	1.52

Notes: Entries report annualized contributions, in percentage points, to sectoral labor productivity growth over 1950–1980, where sectoral labor productivity is $\mathcal{A}_j = Y_j/L_j^p$. The decomposition follows Table 3. Components may not sum exactly to the reported totals because of rounding.

Table H.2. Decomposition of Sectoral Labor Productivity Growth, 1980–2023

	Sector		
	Goods	RI-Serv.	NRI-Serv.
I. Endogenous innovation	2.59	1.32	0.52
(a) <i>Idea growth</i>	2.89	1.79	0.58
(b) <i>Elasticity drift</i>	−0.30	−0.47	−0.06
II. Embodied technical progress	0.07	0.93	0.22
(a) <i>Machine productivity</i>	−0.06	0.62	0.10
(b) <i>Machine market power</i>	0.13	0.31	0.12
III. Relative price effect	−0.66	−0.13	0.15
(a) <i>Output price</i>	−0.41	−0.21	0.15
(b) <i>Sectoral market power</i>	−0.25	0.07	−0.01
IV. Labor efficiency	0.38	0.41	0.39
Total	2.37	2.53	1.27

Notes: Entries report annualized contributions, in percentage points, to sectoral labor productivity growth over 1980–2023, where sectoral labor productivity is $\mathcal{A}_j = Y_j/L_j^p$. The decomposition follows Table 3. Components may not sum exactly to the reported totals because of rounding.

H.2 Non-Homothetic Demand, Relative Prices, and Markups

The main text reports the aggregate productivity implications of shutting down income effects, relative-price effects, and markup drift. This appendix reports the corresponding cumulative accounting results for sectoral allocations, innovation, prices, and productivity.

Because setting $\sigma = 1$ removes both the expenditure-share response to relative prices and the income channel, the income and relative-price shutdowns must be applied in that order. We therefore treat these two channels as an ordered block and average marginal contributions across the two admissible sequences: income, relative prices, then markups; and markups, income, then relative prices. The intermediate columns below sequentially remove these average contributions from the baseline.

Tables H.3 and H.4 report the detailed sectoral outcomes underlying the aggregate results discussed in Section 5.3.

Table H.3. Accounting for Structural Change: Sectoral Allocations

	Baseline		Income		Relative Prices		Markups	
	1950	2023	1950	2023	1950	2023	1950	2023
<i>Panel A. Sectoral value-added shares</i>								
Goods	0.414	0.181	0.414	0.236	0.414	0.456	0.414	0.414
RI-Services	0.074	0.180	0.074	0.078	0.074	0.070	0.074	0.074
NRI-Services	0.512	0.639	0.512	0.686	0.512	0.475	0.512	0.512
<i>Panel B. Sectoral innovation employment shares</i>								
Goods	0.688	0.311	0.714	0.479	0.764	0.816	0.799	0.819
RI-Services	0.048	0.376	0.033	0.120	0.029	0.054	0.016	0.106
NRI-Services	0.264	0.312	0.252	0.401	0.208	0.130	0.185	0.075
<i>Panel C. Sectoral production employment shares</i>								
Goods	0.429	0.188	0.429	0.244	0.429	0.487	0.429	0.530
RI-Services	0.069	0.128	0.069	0.058	0.069	0.050	0.069	0.038
NRI-Services	0.502	0.684	0.502	0.699	0.502	0.463	0.502	0.432

Notes: Baseline reports equilibrium outcomes with all three channels active. To account for interactions across channels, we shut down the income, relative-price, and markup channels in each admissible order and compute the marginal contribution of each channel in each sequence. Because the relative-price shutdown also eliminates the income channel, income must precede relative prices. Starting from Baseline, the Income column removes the average contribution of the income channel, the Relative Prices column additionally removes the average contribution of the relative-price channel, and the Markups column additionally removes the average contribution of markup drift and therefore equals the full-shutdown equilibrium. Panel A reports sectoral value-added shares, while Panels B and C report sectoral shares of innovation and production employment, respectively. The aggregate share of employment devoted to innovation is omitted because it follows the same externally calibrated path in all exercises and is therefore unchanged across specifications. RI- and NRI-Services denote research- and non-research-intensive services, respectively.

Table H.4. Accounting for Structural Change: Productivity, Innovation, Prices, and Output

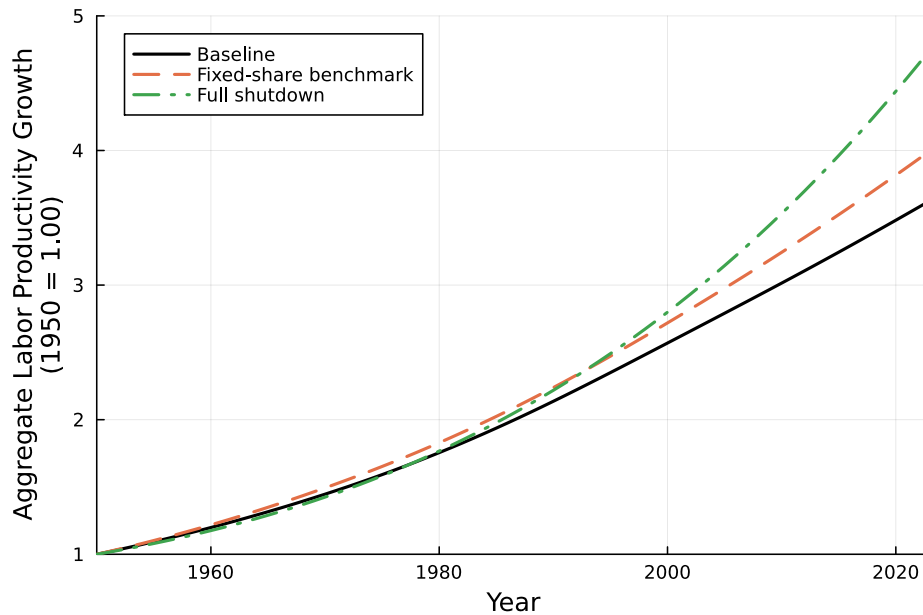
	Baseline	Income	Relative Prices	Markups
<i>Panel D. Labor productivity indices (1950 = 1)</i>				
Goods	6.058	7.242	9.110	10.037
<i>Average annual growth (%)</i>	2.5	2.7	3.0	3.2
RI-Services	5.593	5.198	5.222	5.601
<i>Average annual growth (%)</i>	2.4	2.3	2.3	2.4
NRI-Services	2.720	2.880	2.852	2.698
<i>Average annual growth (%)</i>	1.4	1.5	1.4	1.4
Aggregate	3.627	3.796	4.509	4.747
<i>Average annual growth (%)</i>	1.8	1.8	2.1	2.1
<i>Panel E. Machine-variety indices (1950 = 1)</i>				
Goods	7.509	9.189	11.534	11.730
RI-Services	6.477	4.705	3.919	3.942
NRI-Services	1.521	1.565	1.401	1.365
<i>Panel F. Sectoral price indices (1950 = 1)</i>				
Goods	0.583	0.562	0.563	0.390
RI-Services	0.923	1.128	1.418	1.552
NRI-Services	1.280	1.380	1.783	2.084
<i>Panel G. Aggregate output index (1950 = 1)</i>				
Output	7.699	8.345	9.917	9.968
<i>Average annual growth (%)</i>	2.8	2.9	3.1	3.1

Notes: The table continues the decomposition reported in Table H.3. Starting from Baseline, the Income column removes the average marginal contribution of the income channel, the Relative Prices column additionally removes the average marginal contribution of the relative-price channel, and the Markups column additionally removes the average marginal contribution of markup drift and therefore equals the full-shutdown equilibrium. The marginal contributions are averaged across the two admissible orders described in the notes to Table H.3. Average annual growth rates are annualized log changes in percent. Aggregate labor productivity is real value added per production worker, $VA/L^p = C/L^p$. Machine-variety, sectoral-price, and aggregate-output indices are normalized to one in 1950. RI- and NRI-Services denote research- and non-research-intensive services, respectively.

H.3 Aggregate Labor Productivity Paths

Figure H.1 reports the cumulative aggregate labor productivity paths corresponding to the baseline, the fixed-share accounting benchmark, and the full-shutdown economy discussed in Section 5.3. The figure complements the endpoint results reported in the main text by showing how the differences accumulate over the sample.

Figure H.1. Aggregate Labor Productivity, 1950–2023



Notes: The figure reports cumulative aggregate labor productivity indices over 1950–2023, normalized to one in 1950. Aggregate labor productivity is real value added per production worker, $VA/L^p = C/L^p$. The baseline allows structural change and markup drift. The fixed-share benchmark aggregates baseline sectoral productivity growth using 1950 value-added shares. The full-shutdown counterfactual sets $\eta_j = 1$ and $\sigma = 1$ and holds final-producer and machine-producer markups at their 1950 values.

I Calibration Strategy

We use a nested block-recursive simulated method of moments (SMM) routine to calibrate the internally determined parameter vector

$$\Theta = \{\sigma, \theta_m, \theta_{rs}, \eta_m, \eta_{rs}, \kappa_m, \kappa_{rs}, \kappa_{ns}, \zeta_m(0), \zeta_{rs}(0), \zeta_{ns}(0), g_{\zeta,m}, g_{\zeta,rs}, g_{\zeta,ns}\}.$$

We normalize $\eta_{ns} = 1$ and recover $\theta_{ns} = 1 - \theta_m - \theta_{rs}$.

The calibration of the innovation block uses information on both the allocation and accumulation of innovation. Sectoral scientist shares are particularly informative about relative research productivity across sectors, but do not by themselves determine its common level and trend. We therefore use Goods as the anchor. For Goods, we jointly use the observed patent-stock and labor-productivity paths. The patent-stock path disciplines the accumulation of machine varieties, while the labor-productivity path disciplines the productivity contribution of those varieties. Together, these moments discipline the Goods love-for-variety parameter, κ_m , the initial level of Goods research productivity, $\zeta_m(0)$, and its growth rate, $g_{\zeta,m}$. Conditional on this Goods anchor, the RI-Services and NRI-Services research-productivity parameters are disciplined by their sectoral scientist-share paths. Because knowledge accumulation links these objects across sectors, the corresponding parameter blocks are iterated to a fixed point.

The routine consists of five steps:

1. **Initialize the parameter vector.** Normalize $\eta_{ns} = 1$ and use $\theta_{ns} = 1 - \theta_m - \theta_{rs}$. Choose starting values for

$$\{\sigma, \eta_m, \eta_{rs}, \kappa_m, \kappa_{rs}, \kappa_{ns}, \zeta_m(0), \zeta_{rs}(0), \zeta_{ns}(0), g_{\zeta,m}, g_{\zeta,rs}, g_{\zeta,ns}\}.$$

2. **Recover consumption weights and relative-price wedges.** Given the remaining parameters, recover θ_m and θ_{rs} analytically so that the model matches the 1950 value-added shares of Goods and RI-Services, respectively. The NRI-Services weight then follows from $\theta_{ns} = 1 - \theta_m - \theta_{rs}$. We similarly recover the sectoral relative-price wedges $\tau_{rs}^p(t)$ and $\tau_{ns}^p(t)$ period by period so that the model reproduces the observed relative-price paths.
3. **Calibrate the innovation and love-for-variety parameters.** Conditional on the current demand parameters, proceed as follows:
 - (a) Given candidate values of κ_{rs} and κ_{ns} , jointly calibrate

$$\{\kappa_m, \zeta_m(0), g_{\zeta,m}\}$$

by SMM using the Goods patent-stock and labor-productivity paths over 1950–2023. The patent-stock path provides information on the accumulation of Goods machine varieties, while the labor-productivity path provides information on the productivity contribution of those varieties. Jointly, these paths discipline κ_m , $\zeta_m(0)$, and $g_{\zeta,m}$.

- (b) Conditional on the Goods block $\{\kappa_m, \zeta_m(0), g_{\zeta,m}\}$, jointly calibrate

$$\{\zeta_{rs}(0), \zeta_{ns}(0), g_{\zeta,rs}, g_{\zeta,ns}\}$$

by SMM to minimize the discrepancy between the model and data for the RI-Services and NRI-Services scientist-share paths over 1950–2023. These paths discipline research productivity in the two service sectors relative to the Goods anchor.

- (c) Repeat parts (a) and (b) until κ_m and the six research-productivity parameters converge to a fixed point.
- (d) Given this fixed point, compute the residuals between the model and data for average labor productivity growth in RI-Services and NRI-Services over 1950–2023. If either residual exceeds the specified tolerance, update κ_{rs} and κ_{ns} using a root solver and repeat parts (a)–(d).
- (e) Continue until the model matches the RI-Services and NRI-Services labor-productivity growth rates within the specified tolerance.
4. **Calibrate the demand parameters.** Hold the love-for-variety and research-productivity parameters obtained in Step 3 fixed. Calibrate $\{\eta_m, \eta_{rs}, \sigma\}$ by SMM, minimizing the discrepancy between the model and data for the annual Goods and RI-Services value-added-share paths and the annual Goods relative-price path, normalized to one in 1950.
5. **Iterate across the innovation and demand blocks.** Repeat Steps 3 and 4 until the twelve numerically calibrated parameters

$$\{\sigma, \eta_m, \eta_{rs}, \kappa_m, \kappa_{rs}, \kappa_{ns}, \zeta_m(0), \zeta_{rs}(0), \zeta_{ns}(0), g_{\zeta,m}, g_{\zeta,rs}, g_{\zeta,ns}\}$$

converge to a fixed point. The analytical recovery of

$$\{\theta_m, \theta_{rs}, \theta_{ns}, \tau_{rs}^p(t), \tau_{ns}^p(t)\}$$

described in Step 2 is performed each time the model is solved.

J Proofs

J.1 Proof of Proposition 1

Fix a sector j and suppress time arguments throughout. In the symmetric equilibrium, all machine varieties are used in the same quantity, $x_j(\nu) = x_j$. The machine aggregator in Equation (4) therefore becomes

$$X_j = N_j^{\kappa_j} \left(\int_0^{N_j} x_j^{\mu_j^x} d\nu \right)^{1/\mu_j^x} = N_j^{\kappa_j + \frac{1}{\mu_j^x}} x_j.$$

The first-order conditions of the variety producer imply

$$p_j^X X_j = \mu_j^y (1 - \alpha) p_j Y_j.$$

Under symmetry, total expenditure on machine varieties is

$$p_j^X X_j = N_j p_j^x x_j.$$

Using the optimal machine price from Equation (9),

$$p_j^x = \frac{1}{\mu_j^x z_j},$$

we obtain

$$x_j = \mu_j^y \mu_j^x (1 - \alpha) z_j \frac{p_j Y_j}{N_j}.$$

Substituting this expression into the machine aggregator gives

$$X_j = \mu_j^y \mu_j^x (1 - \alpha) p_j z_j Y_j N_j^{\kappa_j + \frac{1 - \mu_j^x}{\mu_j^x}}.$$

Using the variety-level production technology,

$$Y_j = X_j^{1-\alpha} (\bar{h}_j L_j^p)^\alpha,$$

we obtain

$$Y_j = \left[\mu_j^y \mu_j^x (1 - \alpha) p_j z_j Y_j N_j^{\kappa_j + \frac{1 - \mu_j^x}{\mu_j^x}} \right]^{1-\alpha} (\bar{h}_j L_j^p)^\alpha.$$

Collecting terms in Y_j yields

$$Y_j^\alpha = [\mu_j^y \mu_j^x (1 - \alpha) p_j z_j]^{1-\alpha} N_j^{\left(\kappa_j + \frac{1 - \mu_j^x}{\mu_j^x}\right)(1-\alpha)} (\bar{h}_j L_j^p)^\alpha.$$

Raising both sides to the power $1/\alpha$ and dividing by L_j^p gives

$$\frac{Y_j}{L_j^p} = N_j^{\left(\kappa_j + \frac{1-\mu_j^x}{\mu_j^x}\right)\left(\frac{1-\alpha}{\alpha}\right)} \left[\mu_j^y \mu_j^x (1-\alpha) p_j z_j\right]^{\left(\frac{1-\alpha}{\alpha}\right)} \bar{h}_j.$$

Defining

$$e_j^Y \equiv \left(\kappa_j + \frac{1-\mu_j^x}{\mu_j^x}\right) \left(\frac{1-\alpha}{\alpha}\right)$$

delivers Equation (11). □

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